

Untangling key abiotic predictors of terrestrial mammal diversity patterns across ecoregions and species groups in Kenya

Kenneth Otieno Onditi^{a,b,c,f}, Wenyu Song^{a,g}, Xueyou Li^a, Simon Musila^{c,e},
Zhongzheng Chen^{a,d}, Quan Li^a, James Mathenge^e, Esther Kioko^{c,f}, Xuelong Jiang^{a,f,*}

^a State Key Laboratory of Genetic Resources and Evolution, Kunming Institute of Zoology, Chinese Academy of Sciences, Kunming, China

^b Kunming College of Life Sciences, University of Chinese Academy of Sciences, Kunming, China

^c Department of Zoology, National Museums of Kenya, Nairobi, Kenya

^d Collaborative Innovation Center of Recovery and Reconstruction of Degraded Ecosystem in Wanjiang Basin Co-funded by Anhui Province and Ministry of Education, School of Ecology and Environment, Anhui Normal University, Anhui, China

^e Kenya Wildlife Services, Nairobi, Kenya

^f Sino-Africa Joint Research Centre, Chinese Academy of Sciences, Nairobi, Kenya

^g Vector Laboratory, Institute of Pathogens and Vectors, Yunnan Provincial Key Laboratory for Zoonosis Control and Prevention, Dali University, Dali, China

ARTICLE INFO

Keywords:

Biodiversity conservation
Functional diversity
Human footprint
Kenya
Phylogenetic diversity
Protected areas
Species richness
Terrestrial mammal

ABSTRACT

Understanding the interactions between abiotic (environmental and anthropogenic) factors and species diversity and distribution patterns is fundamental to improving the ecological representativeness of biodiversity management tools such as protected areas (PAs). However, significant knowledge gaps remain about how species' ecological and evolutionary opportunities are associated with abiotic factors, especially in biodiversity-rich but economically ill-equipped countries such as Kenya. Here, we explored the interactions of terrestrial mammal diversity patterns and abiotic factors across species groups and ecoregions in Kenya. We coupled data on terrestrial mammal occurrences, phylogeny, functional traits, and environmental predictors in Kenya to derive multiple diversity indices, encompassing species richness and phylogenetic and functional richness, and mean pairwise and nearest taxon distances. We explored the interactions of these indices with several abiotic factors using multivariate regression analyses while adjusting for spatial autocorrelation. The results showed weak correlations between species richness versus the phylogenetic and functional diversity indices. The best-fit models explained variable proportions of diversity indices between species groups and ecoregions and consistently retained annual temperature and precipitation averages and seasonality and human footprint as the strongest predictors. Compared to the species-poor xeric northern and eastern Kenya regions, the predictors had weak associations with diversity variances in the species-rich mesic western and central Kenya regions, similar to focal species groups compared to ordinal classifications and the combined species pool. These findings illustrate that climate and human footprint interplay determine multiple facets of terrestrial mammal diversity patterns in Kenya. Accordingly, curbing human activities degrading long-term climatic regimes is vital to ensuring the ecological integrity of terrestrial mammal communities and should be integrated into biodiversity management frameworks. For a holistic representation of critical conservation areas, biodiversity managements should also prioritize terrestrial mammal phylogenetic and functional attributes besides species richness.

1. Introduction

An accurate understanding of the factors determining species diversity patterns and processes can help design ecologically-sound biodiversity management strategies (Malhi et al., 2020; Neugarten et al., 2020). Across biodiversity-rich sub-Saharan region countries such

as Kenya, which host a large proportion of the global mammal biodiversity (Brown, 2014), there are hardly national-level investigations into the determinants of the faunal diversity patterns and processes. This hinders the meaningful country-specific integration of ecological considerations into biodiversity management policies since conservational initiatives are dominantly nationally biased and associations between

* Corresponding author at: Kunming Institute of Zoology, Chinese Academy of Sciences, 17 Longxin Road, Kunming, Yunnan 650201, China.

E-mail address: jiangxl@mail.kiz.ac.cn (X. Jiang).

<https://doi.org/10.1016/j.ecolind.2023.110595>

Received 3 January 2023; Received in revised form 4 May 2023; Accepted 28 June 2023

Available online 5 July 2023

1470-160X/© 2023 The Authors. Published by Elsevier Ltd. This is an open access article under the CC BY license (<http://creativecommons.org/licenses/by/4.0/>).

biodiversity patterns against human pressure are highly variable across countries (Shi et al., 2005). Overall, the compounded effects of the increasingly intensifying human activity pressure in wildlife areas, unpredictable climate regimes, and global warming on biodiversity is a developing discussion that requires more studies to understand the impacts better, predict the ecosystem and species responses, and effectively manage detrimental influences.

Systematic variabilities in the associations between faunal diversity patterns and abiotic factors suggest that species synergistically respond to ecological and evolutionary opportunities non-randomly (Ricklefs, 2004; Rowan et al., 2020). Nonetheless, a unifying theory of how abiotic factors structure faunal biodiversity patterns in tropical regions remains debated due to complex deterministic and stochastic mechanisms simultaneously driving species' diversity and distribution patterns (Ferrer-Castán et al., 2016; Vetaas, 2006; Woodbridge et al., 2021). Under these complex diversity-predictor associations, patterns observed in combined species pools or at large spatial scales might poorly surrogate those of individual clades, guilds, or particular geographies. Thus, while water-energy dynamics strongly affect mammal diversity patterns in tropical regions (Hawkins et al., 2003), more intricate interactions may define species persistence at local scales, between regions, and smaller species groups. Furthermore, Kenya's complex geophysical environment spanning coastal forests, alpine mountain moorlands, lowland savanna grassland-woodland mosaics, sky-island Afromontane forests, and predictable climatic regimes implies that the mammalian diversity patterns are potentially determined by multiple abiotic variables, leading to more complex diversity-predictor associations (Mukeka et al., 2019; Onditi et al., 2021; Onditi et al., 2022; Tyrrell et al., 2020; Western et al., 2015; Zhang et al., 2022).

Three factors have emerged as the most significant determinants of mammal diversity and distribution patterns in the tropics — climate, habitat heterogeneity, and human footprint. Firstly, the tropics' stable climate (temperature and precipitation regimes) is systematically strongly associated with mammalian diversity patterns (Brown, 2014; Fine, 2015; Hawkins et al., 2003), pertaining to the positive feedback between stable climates and resource availability. Under stable climates (climatic stability hypothesis), species can develop finer adaptations within narrow niches, leading to higher speciation rates and consequent higher species abundance and diversity (Brown, 2014; Erwin, 2009). While climatic variables (water and energy availability) are dominantly positively associated with species richness at global and regional scales, the strength of the associations may be blurred at local scales and among lower taxonomic groups (Ferrer-Castán et al., 2016). How Kenya's nominally variable and predictable climate affects terrestrial mammals' diversity needs further investigation.

Secondly, habitat heterogeneity from vegetation structure and topography has also emerged as a strong driver of faunal diversity patterns in tropical regions (Allouche et al., 2012; Kerr and Packer, 1997; Thomsen et al., 2022). Under the habitat heterogeneity hypothesis, terrestrial mammal abundance and diversity increase with increasing habitat heterogeneity (MacArthur and MacArthur, 1961). The positive interactions result from the higher productivity of more complex vegetation structures and topographies offering more abundant and diverse niche opportunities, leading to more diverse and dense mammal assemblages (Couvreur et al., 2021; Gorczynski et al., 2021; McCain et al., 2018). Topography (elevation and terrain ruggedness), per se, controls key hydrological and temperature processes that determine habitat distribution and quality for terrestrial mammals and directly influence species distribution by manipulating movement, occurrence, predator–prey relationships, functional trait adaptations, and niche dimensions (Amatulli et al., 2018; Stein and Kreft, 2015). Also, harsh habitats (characterized as xeric regions with high annual temperature and low annual precipitation) have stronger filtering effects, where related species tend to share more convergent trait(s) that better adapt them to the prevailing environmental conditions, leading to trait(functional) clustering. Trait divergence, on the other hand, is characteristic

of species in benign habitats having weaker environmental filtering effects, which leaves competitive interactions (limiting similarity dynamics) as the stronger diversity predictor (Cornwell et al., 2006; Kang et al., 2016; Song et al., 2020; Weiher et al., 1998). To better account for regionalization in species composition due to habitat heterogeneity, there is a need for more studies focused on how vegetation structure and topography drive terrestrial mammal species richness and phylogenetic and functional diversity, more so on variabilities between species groups and ecoregions.

Thirdly, anthropogenic influence (human activity pressure) consistently emerges as a vital driver of terrestrial mammal diversity patterns (Andermann et al., 2020; Barrett et al., 2011; Ceballos et al., 2015; Lindsey et al., 2022; Mukeka et al., 2019; Ogutu et al., 2016; Veldhuis et al., 2019; Li et al., 2021), under the intermediate disturbance hypothesis, which postulates optimum diversity at intermediate disturbance intensities (Connell, 1978). There is increasing ubiquity in consensus that human activities dominantly negatively impact wildlife diversity and ecology, with the impacts more severe in biodiversity-rich tropical countries, where the subsistence activities of most people (agriculture), human settlement, and urbanization contend for land use with wildlife (Barrett et al., 2011; Mukeka et al., 2019; Ogutu et al., 2016; Veldhuis et al., 2019). Furthermore, the attribution of recent increases in animal biodiversity declines and major historical extinction events to human activity impacts on ecosystems (Andermann et al., 2020; Ceballos et al., 2015) calls for a more informed understanding of how human activities influence terrestrial mammal diversity. Kenya's fast-paced human population growth and urbanization drive human encroachment into previously relatively intact wildlife (Lindsey et al., 2022); the exact implications of these ecological footprints on biodiversity conservation derive from complex spatiotemporally-dependent interactions which need more studies to understand and manage better.

Here, we investigated the association patterns between multiple terrestrial mammal diversity indices and abiotic (environmental and human) factors in Kenya to provide a comprehensive national-level eco-evolutionary insight into the nexuses between diversity-environment interactions and biodiversity conservation. Our focus on terrestrial mammals pertains mainly to their high impact on national biodiversity management performances in Kenya (Onditi et al., 2021) and ecological effects on ecosystems (Lacher et al., 2019). The main objectives were to (i) disentangle the main abiotic predictors (environmental and human) of terrestrial mammal diversity patterns in Kenya and (ii) unravel the pattern of effects of environmental and human factors on terrestrial mammal diversity between species groups (clades and guilds) and geographies (ecoregions). We adopted a hypothetico-deductive model to test three hypotheses most cited in explaining vertebrate diversity patterns in tropical systems such as the Afrotropical realm (Brown, 2014; Couvreur et al., 2021; Gorczynski et al., 2021; Rowan et al., 2020). These included the water-energy (Rahbek, 1995; Rosenzweig, 1992), habitat heterogeneity (McCain et al., 2018; Rosenzweig, 1992; Rosenzweig, 1995), and intermediate disturbance (Connell, 1978) hypotheses. For the first objective, we broadly expected strong associations between diversity patterns and water-energy variables and human footprint, with species richness increasing as water availability and habitat heterogeneity increase, decreasing as temperature increases (Gillman and Wright, 2014; Hawkins et al., 2003; Li et al., 2013; Luo et al., 2012), and have a hump-shaped association with human footprint (Cazalis, 2022), while the phylogenetic and functional diversity indices would portray variably contrasting associations (Le Bagousse-Pinguet et al., 2019). For the second objective, we hypothesized variable diversity-predictor associations between species groups and ecoregions due to intimately variable associations between species' evolutionary history and ecological opportunities, with higher explained diversity variances in harsh than in benign ecoregions (Brown, 2014; Kissling et al., 2012; Montaña-Centellas et al., 2020; Montaña-Centellas et al., 2022; Onditi et al., 2022; Ramesh et al., 2016; Rowan et al., 2020; Rowe, 2009).

2. Material and Methods

See an extended account of Material and Methods in [Supplementary material: Table S1](#).

2.1. Kenya's terrestrial mammal occurrence records

We extracted accounts of Kenya's protected areas (PAs) from [UNEP-WCMC and IUCN \(2021\)](#). After dropping PAs with no land overlaps and spatially overlapping ones, we retained 400 PAs for consequent analyses. To derive species ranges/occurrences, we used the species checklist of [Musila et al. \(2019\)](#) as a backbone taxonomy of terrestrial mammals in Kenya and matched them with species' range/occurrence records from IUCN ([UNEP-WCMC and IUCN, 2021](#)) and GBIF ([GBIF.org, 2020](#)). To ensure that taxonomic accounts and range representations were the most up-to-date, we validated the extracted range extents against the distribution accounts of [Musila et al. \(2019\)](#) and regional and global species checklists ([Burgin et al., 2020](#); [Kingdon et al., 2013](#); [Mittermeier et al., 2013](#)). The ranges of species currently known to be occurring in Kenya from these checklists but missing from the IUCN and GBIF databases were appended by referring to their literature distribution accounts to map and draw new range polygons.

From the combined records, we compiled two species occurrence datasets (community datasets) — a 400 PAs-level and the nationwide grids-level community-by-species matrices. We estimated the presence-absence species occurrences into lists of all terrestrial mammal species in every PA and grid, joining species ranges to grids and PAs based on whether the layers intersected or overlapped, resulting in lists of all species present in PAs or grids, which we then converted into PA/grid-by-species matrices. We defined the grids following preliminary exploratory analyses, for which Kenya's administrative extent was divided into 2,109 300-km² hexagonal grids. Because mammalian distributions and responses to environmental conditions are geographically variable, we used species composition dissimilarity and phylogenetic beta diversity ([Daru et al., 2020](#)) to classify PAs and grids into clusters [hereon just *ecoregions*]. First, we worked out species composition dissimilarity distances based on the Jaccard similarity index, translated the distances into ultrametric dendrograms based on the average linkage hierarchical clustering algorithm, and 'cut' the dendrogram into pre-defined(5) groups — *ecoregions*.

2.2. Community taxonomic, phylogenetic, and functional characterization

2.2.1. Data collection

To account for variabilities in assembling patterns due to ancestry and ecological functions ([Zupan et al., 2014](#)), the species checklist was apportioned into ten groups comprising three focal species groups (large carnivores, large herbivores, and endangered species; [Onditi et al. \(2021\)](#)), six ordinal classes with ≥ 5 species (Carnivora, Cetartiodactyla, Chiroptera, Eulipotyphla, Primates, and Rodentia), and a combined species pool.

The input phylogeny for estimating phylogenetic diversity was reconstructed from [Kumar et al. \(2017\)](#)'s mega tree. We used the checklist of Kenya's terrestrial mammals to prune the phylogeny into only species in the community dataset, with species missing from the extracted phylogeny randomly assigned to the respective genera (representing the generic kinship's most recent common ancestor).

The input trait data for estimating functional diversity indices were represented by external body characters, diet, and activity patterns; the data were extracted from [Wilman et al. \(2014\)](#) and [Jones et al. \(2009\)](#). The specific traits included body mass, ten categories of species diet proportion, and three levels of activity patterns ([Supplementary material: Table S2](#)). To handle distribution normality and harmonize the traits' weights, we variously Log₁₀-transformed body mass and Principal component analysis [PCA]-transformed diet traits; the first five PCA

axes, which together explained 97.84% variance as new diet traits. Traits were standardized by scaling to zero mean and unit variance before analyses.

2.2.2. Diversity indices

Biodiversity encompasses multiple attributes, including species richness and abundance, and different evolutionary lineages and ecological strategies that are not necessarily interrelated, such that surrogacy between taxonomic diversity, phylogenetic, and functional diversity can be relatively poor ([Pavoine and Bonsall, 2011](#); [Thuiller et al., 2006](#); [Tucker et al., 2017](#)). Therefore, we selected a priori 12 diversity indices encompassing taxonomic, phylogenetic, and functional richness (sum of differences among observations), divergence (average differences among observations), and regularity/structure (variance of observations). For richness, we estimated species richness from the incidence-based taxonomic alpha diversity (SR), phylogenetic richness using Faith's phylogenetic diversity index (PD) ([Faith, 1992](#)), and functional richness using the same method as PD after converting the species traits matrix to a dendrogram (FD). We represented divergence indices using the richness-controlled phylogenetic and functional mean pairwise and mean nearest taxa distances. These included phylogenetic mean pairwise distance (PD^{MPD}) and mean nearest taxon distance (PD^{MNTD}) and functional mean pairwise distance (FD^{MPD}) and mean nearest taxon distance (FD^{MNTD}). The mean pairwise distances capture the mean pairwise distances between all species in a community and correspond to tree basal-weighted dynamics, while the nearest taxa distances capture pairwise distances between relatives and correspond to tree tip/tip-weighted dynamics ([Webb et al., 2002](#); [Webb and Donoghue, 2005](#)). For regularity/structure, we estimated the standardized effect sizes of the divergence dimensions — phylogenetic and functional community structure based on nearest taxon indexes (PD^{NTI} and FD^{NTI}, respectively) and nearest relative indexes (PD^{NRI} and FD^{NRI}, respectively), by comparing the observed divergence distances to null-model communities generated from 1000 iterations through 999 randomizations based on the independent swap algorithm ([Gotelli and Entsminger, 2003](#)). In addition, we estimated phylogenetic endemism (hereon just 'endemism') as the sum of 'unique' branch lengths connecting species within communities, with each branch weighted by the fraction of its geographic range in the community. Endemism is an important complementary metric to the above indices vital to better prioritizations of biodiversity conservation, with areas of high endemism key to understanding evolutionary processes behind patterns in biodiversity distributions ([Murali et al., 2021](#)).

2.3. Abiotic drivers of terrestrial mammal diversity patterns

A total of 30 abiotic variables ([Supplementary material: Table S3](#)) representing water-energy dynamics, habitat productivity, topography, and human influence were used as potential diversity predictors based on their observed roles in structuring terrestrial mammal assembling in tropical regions in the literature. The water-energy hypotheses were explored using 19 bioclimatic variables from the CHELSA climate dataset ([Karger et al., 2017](#)). These bioclimatic variables can independently represent water and energy availability for terrestrial mammals within communities and have been widely used to model species distribution patterns and habitat suitability toward better understanding diversity-environment associations for improved biodiversity conservation management. We also incorporated additional energy availability variables, including actual annual evapotranspiration (AET), potential evapotranspiration (PET), and Global Aridity Index (GAI) to capture better the water-temperature dynamics of energy available for plant growth, the fundamental food resource for terrestrial mammals. The evapotranspiration was extracted from MOD16A3GF Version 6 ([Running et al., 2019](#)), while PET and GAI were extracted from [Zomer and Trabucco \(2022\)](#). The habitat heterogeneity hypothesis was explored using habitat productivity and topographic influences. Habitat

productivity was represented by normalized difference vegetation index (vegetation) and enhanced vegetation index (EVI); both variables were extracted from [Didan \(2015\)](#). Topographic influences were represented by elevation and terrain ruggedness index (terrain ruggedness); elevation was extracted from the NASADEM Merged DEM Global 1 arc second V001 ([NASA JPL, 2020](#)), while terrain ruggedness was calculated from the elevation layer using QGIS's *Terrain Ruggedness Index* function. The intermediate disturbance hypothesis was explored based on the influences of human activities on terrestrial mammal diversity — human footprint; the cumulative human influence on terrestrial mammal diversity ([Venter et al., 2016; Venter et al., 2018](#)).

Because variables were extracted as GIS raster files from global coverages, they were first masked to Kenya's extent, resampled to the same spatial resolution (0.00138889 degrees, = 5 arc-seconds) in Arc-Map using the Arc Toolbox (based on the Bilinear interpolation) to enhance spatial resolution and comparability, and then log-transformed (natural logarithms) to normalize distributions — handle skewness. The mean values per grid and PA were obtained using the zonal statistics tool in QGIS, which estimates central tendency measures from the input raster layers corresponding to each grid/PA overlay polygon. Following further literature review and procedures to ensure that the predictors did not violate the assumptions of the regression models, we used Pearson's Correlation Index to identify highly correlated variable sets from which we only retained a single variable. This was implemented for the combined datasets (nationwide) of PAs and grids and individual ecoregions because the correlations highly varied across geographies. Although these predictor variables were extracted from global-resolution datasets, they capture complex interactions between ecological processes and abiotic variables better, thus often perform better than those extracted from local resolutions ([Hallett et al., 2004](#)).

2.4. Multivariate analysis of diversity-driver associations

Using distance-based redundancy analysis, we first explored the drivers of species composition between PAs and grids ([Legendre and Anderson, 1999](#)) and then investigated the associations between the diversity indices and abiotic predictors using multivariate regression analysis.

We performed variables' selection prior to regression analyses to avoid redundancy and inflated standard errors. The selection was based on bivariate correlations where we retained only single variables from highly correlated sets — Pearson correlation coefficient, $r > 0.7$ ([Zurell, 2020](#)) followed by variance inflation factor (VIF) analysis to further check for and handle multicollinearity. For this, the generalized linear model (GLM) was used with a VIF cut-off of < 5 as a threshold at which highly correlated predictors (VIFs > 5) were consecutively dropped, VIFs recalculated, and the procedure replicated until all predictors achieved the threshold. The resulting models were then used for model selection, i.e., to find the 'best-fit model' with the best combination of the retained predictors explaining the observed diversity patterns. Here, the *glmulti* package ([Calcagno, 2019](#)) was used to screen all possible combinations of predictors based on main effects and the *exhaustive screening* option ranked by the Akaike information criterion (AIC). In many instances, there were several contending top best-fit models, suggesting there was no significant change in AIC scores. Therefore, the top models with δAIC score < 2 were averaged using the *MuMIn* package ([Barton, 2019](#)). Model averaging also aids in handling model selection and prediction errors and is particularly useful for ecological data, which is commonly noisy ([Burnham and Anderson, 2004; Dormann et al., 2018](#)). Because of contrasting correlational associations, predictors and model selection were implemented piecewise between ecoregions ([Supplementary material: Fig. S1](#)).

To account for spatial autocorrelation between PAs and grids resulting from uncoupled geographical influences due to the proximity that violates modeling assumptions of residual independence, we used spatially-explicit regression models — generalized linear mixed effects

models (GLMM) and generalized additive mixed effects models (GAMM), where PA and grid centroid geographical coordinates were added as spatial random effects ([Dormann, 2007; Legendre, 1993](#)). Further, to concurrently compare the performance of different regression algorithms commonly used in predicting diversity-environment associations, we also fitted standard generalized linear models (GLM) without controlling for spatial autocorrelation effects. In the GAMMs, we also included longitude-latitude coordinates as correlated covariates (as spatially correlated random effects), like in the GLMMs. Because the diversity indices and predictors fitted normal distributions well, we fitted the regressions with *Gaussian* probability distribution families and linear *identity* link functions. Furthermore, in the PA-level regressions, geospatial size variations were accounted for by adding PA size as a random effect.

In total, we used 1,080 regression models to explore associations between diversity patterns and the abiotic variables, comprising separate models for each of the ten species groups and five ecoregions across the six unique diversity indices. Only the combined species pool was analyzed between ecoregions in both PA- and grid-based analyses, and separate parallel models were used to explore the singular influence of human footprint. Comparisons of AIC and RMSE estimates from 'preliminary models' showed that GLMMs consistently outperformed or were comparable to GAMMs, with both better than GLMs ([Supplementary material: Fig. S2](#)). Thus, to avoid overfitting and refine the results' understandability, we extracted the top two variables with the highest relative importance to model-explained variances in preliminary GLMMs and GAMMs and used them to fit the final models using GLMM. The statistical significances of associations between diversity variances and predictors were estimated from the effect size of explained diversity variances — model performance indicators [adjusted r -squared, $AdjR^2$; Akaike's Information Criteria, AIC; and root mean square error, RMSE], standardized model parameters [coefficients (including the confidence intervals), t statistics, and p values], and the frequency by which predictors had the highest relative contribution to explained variances, f .

3. Results

The PAs and grids each clustered into five ecoregions based on species composition dissimilarities, with the dendrogram-type relationship in the form “(((central-to-southern, southwestern), northern), (southcoast, northcoast)), malkamari);” in the PAs dataset and “(((northwestern, (tanariverbasin, coastal)), northeastern), southwestern));” in the grids dataset ([Fig. 1](#)).

After PCA-reducing the original 12 diversity indices to six unique ones ([Fig. 2A](#)), higher SR (and, by implication, the indices highly correlated with it) was observed in the southwestern diagonal of the country, progressively declining towards the north-eastern ([Fig. 2B; Supplementary material: Fig. S3](#)). Between the PAs ecoregions, SR was highest in *southwestern*, followed by *central2southern*, *southcoast*, *northern*, and *northcoast*, while in the grids' ecoregions, SR was highest in *southwestern*, followed by *coastal*, *northwestern*, *tanariverbasin*, and *northeastern*. In both datasets, endemism was lower in the species-rich central regions and higher in the southwestern and coastal regions. While species-poor areas tended to have larger PD^{MPD} and FD^{MPD} , the community structure indices (PD^{NTI} and FD^{NTI}) depicted contrasting patterns with notable pockets of overdispersion and clustering ([Fig. 2B; Table 1; Supplementary material: Fig. S3–Fig. S4](#)). Associations between diversity dimensions were highly variable between ecoregions, illustrating a pattern where an increase or decline in species richness did not have unidirectional effects on the phylogenetic and functional divergence and community structure indices ([Supplementary material: Fig. S5](#)).

Across the final 1,080 regression models, the most important predictors (based on the frequency, f , by which predictors had the highest relative contribution to $AdjR^2$) included temperature (f : 34), precipitation seasonality (f : 30), human footprint (f : 25), vegetation (f : 25),

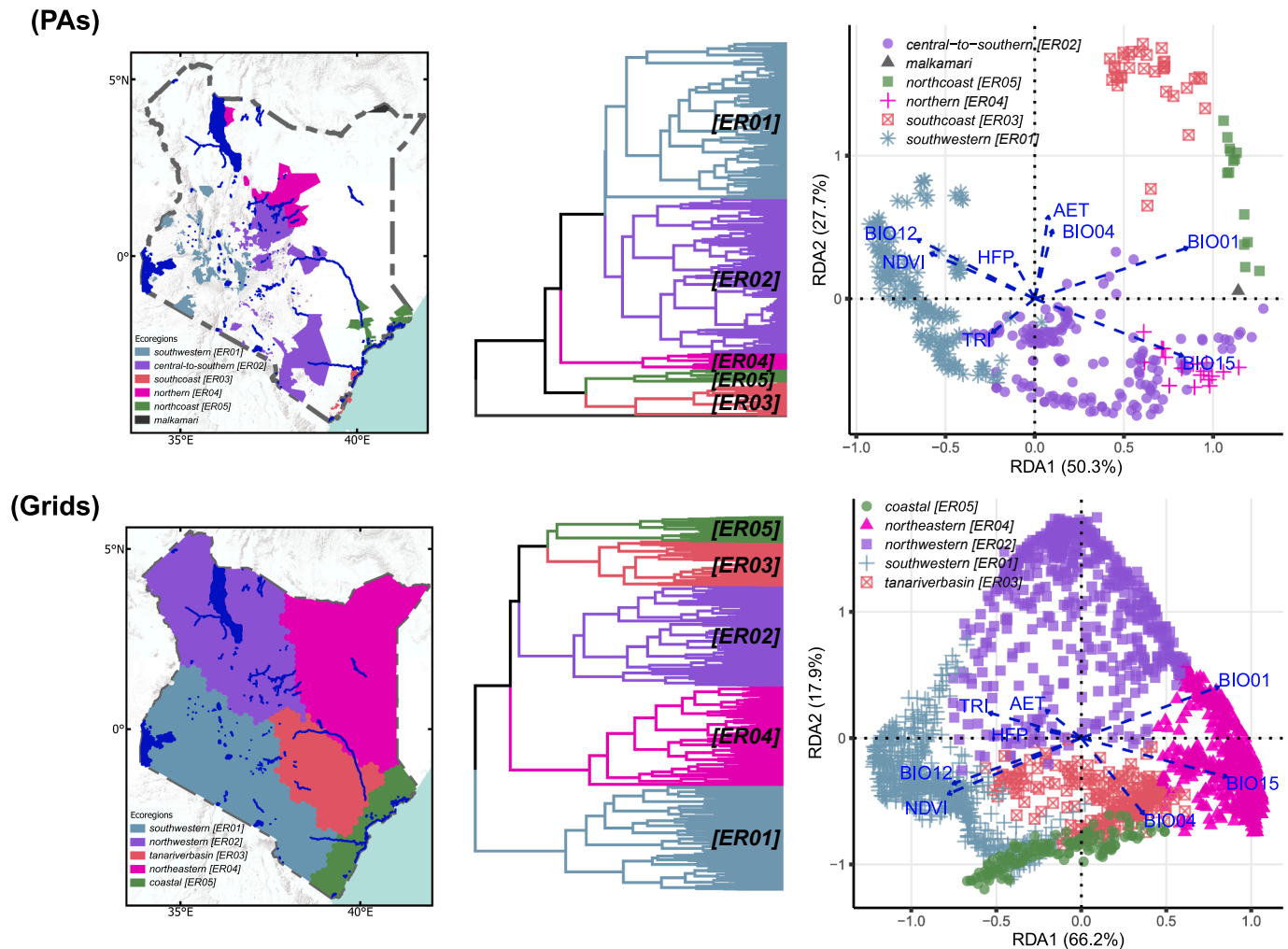


Fig. 1. Terrestrial mammal geographical composition patterns in Kenya. **(PAs)** shows the clustering and regionalization of protected areas [PAs] based on species composition dissimilarity; the figure to the left shows the geographical distribution of Kenya's PAs delimited by *ecoregions*, the middle figure shows the dissimilarity distances as a dendrogram (hierarchical PA associations within *ecoregions*), while the figure to the right shows the ordination plot of the dissimilarity distances based on distance-based redundancy analysis of the influence of abiotic predictors. **(Grids)** shows the scheme of Kenya's terrestrial mammal composition illustrated using the nationwide grids-based dataset. The predictor variables include actual annual evapotranspiration (AET), annual mean temperature (BIO01), annual precipitation (BIO12), normalized difference vegetation index (NDVI), precipitation seasonality (BIO15), temperature seasonality (BIO04), elevation (ELEV), and terrain ruggedness index (TRI).

evapotranspiration (f : 23), precipitation (f : 22), temperature seasonality (f : 17), and terrain ruggedness (f : 4) — [Supplementary material: Fig S6](#).

3.1. Diversity-predictor associations between species groups

The proportion of diversity indices explained by the predictors varied between species groups, with subtle systematic associations between the best predictors and species groups ([Supplementary material: Fig. S5](#)). Across species groups, SR and endemism increased while PD^{MPD} and FD^{MPD} decreased as habitat suitability increased (i.e., when *suitable habitat* is defined as an area of high vegetation and precipitation, low temperature, low temperature and precipitation seasonality, high terrain ruggedness, and low human footprint).

In the PAs dataset, temperature (f = 13), precipitation seasonality (f = 10), human footprint (f = 9), vegetation (f = 9), evapotranspiration (f = 9), precipitation (f = 5), temperature seasonality (f = 4), and terrain ruggedness (f = 1) had the strongest association with diversity variances. In the combined species pool, diversity variances were most strongly correlated with temperature (SR), precipitation seasonality (endemism, PD^{MPD} , and FD^{NTI}), evapotranspiration (FD^{MPD}), and vegetation (PD^{NTI}). On the other hand, among endangered species, human

footprint had the strongest effect on SR, temperature on endemism and FD^{NTI} , temperature seasonality on PD^{MPD} and FD^{MPD} , and precipitation on PD^{NTI} ([Fig. 3](#); [Supplementary material: Table S4](#)). Diversity variances in the carnivore groups (large Carnivores and order Carnivora) were most strongly associated with temperature, precipitation seasonality, and vegetation in the former, while in the latter, evapotranspiration, human footprint, and vegetation were the most important ([Fig. 3](#); [Supplementary material: Table S4](#)). Temperature annual average and seasonality, precipitation seasonality, evapotranspiration, vegetation, and human footprint were the most strongly associated with diversity patterns of herbivores (large herbivores and order Cetartiodactyla) ([Fig. 3](#); [Supplementary material: Table S4](#)).

Like the PAs dataset, there were no systematic associations between individual predictors and species groups or diversity indices in the grids-based analyses. Combined, the predictors with the strongest associations with diversity indices included; precipitation (f = 12), human footprint (f = 11), vegetation (f = 10), precipitation seasonality (f = 8), temperature (f = 7), evapotranspiration (f = 6), temperature seasonality (f = 5), and terrain ruggedness (f = 1). Predictor effects were strongest in Eulipotyphla, Chiroptera, and Rodentia and weakest in Carnivora, Primates, and large carnivores ([Supplementary material: Table S4](#);

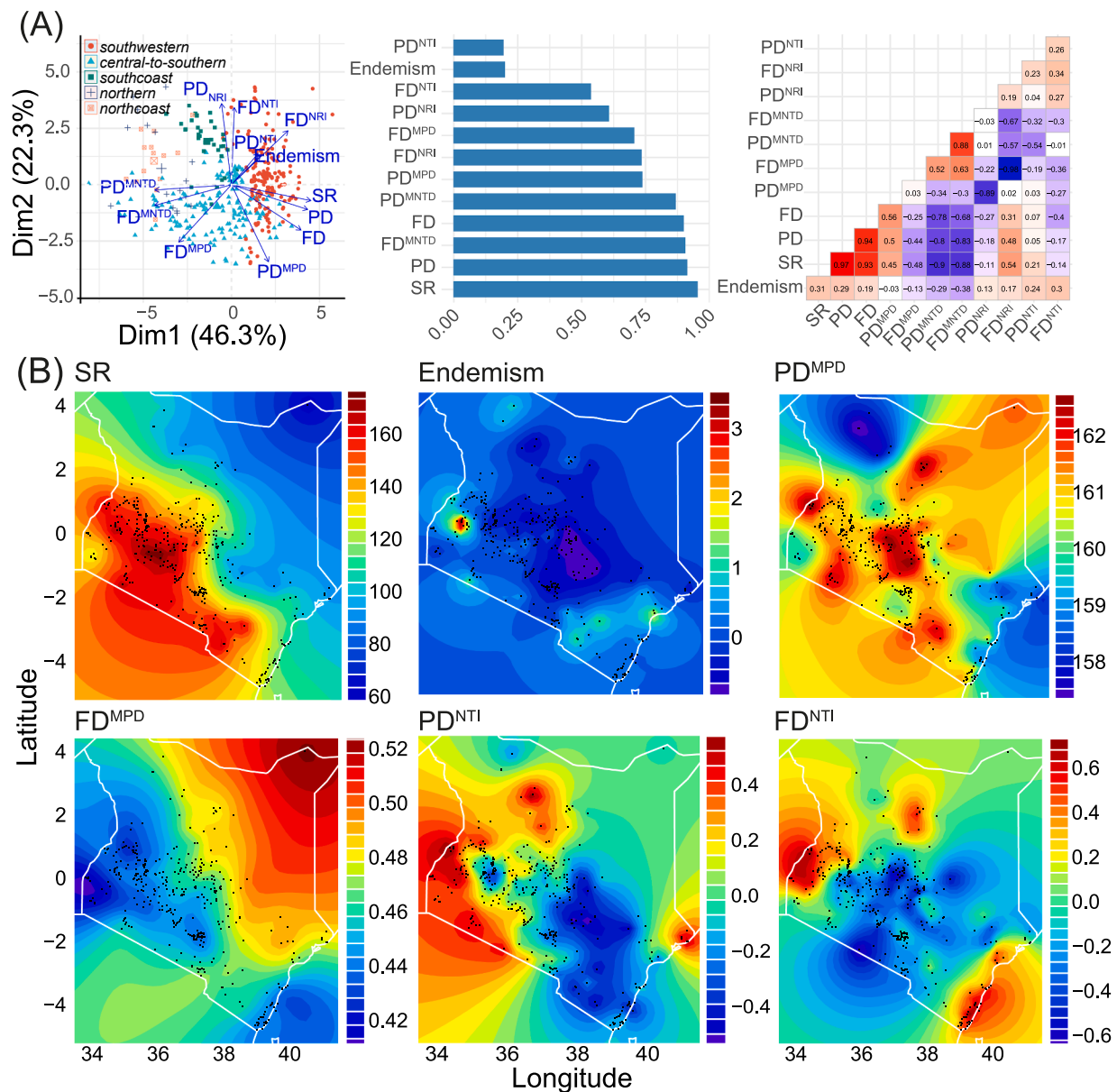


Fig. 2. The terrestrial mammal diversity and distribution patterns in Kenya using multiple diversity indices based on protected areas (PAs). **(A)** shows the correlation between observed diversity patterns based on principal component analysis (PCA); the figure to the left is a biplot of the first two PCA axes, the middle figure is the quality of indices' representation in the PCA, and the figure to the right is indices' bivariate correlations. **(B)** shows the geographical distribution of unique diversity indices (Pearson Bivariate correlation < 0.7) based on spatially-explicit generalized linear mixed models (GLMMs) predictions. The GLMMs refit models based on observed values to predict unobserved values (outside PAs); black dots represent individual PAs, and white lines Kenya's administrative boundary. See [Supplementary material: Fig. S3](#) for the equivalent grid-based plots.

Supplementary material: Fig. S6). In the combined species pool, the explained diversity variances were most strongly contributed to by precipitation (endemism, PD^{MPD}, and FD^{NTI}), precipitation seasonality (SR), temperature (FD^{MPD}), and evapotranspiration (PD^{NTI}). These variables also overlapped as the strongest predictors of diversity variances in the endangered species group — precipitation (endemism), precipitation seasonality (PD^{NTI}), temperature (FD^{MPD}), vegetation (PD^{MPD} and FD^{NTI}), and human footprint (SR) (Fig. 3; [Supplementary material - Fig. S7](#); [Supplementary material: Table S4](#)). The diversity-predictor variations in the combined species pool were mirrored in the rest of the species groups. For instance, among the large mammals (large carnivores and large herbivores), diversity variances had the strongest association with precipitation (FD^{NTI}), precipitation seasonality (SR and FD^{MPD}), temperature seasonality (PD^{MPD} and PD^{NTI}), and vegetation (endemism) in large carnivores. In large herbivores, diversity variances

were mainly affected by precipitation (FD^{NTI}), temperature (endemism), temperature seasonality (PD^{NTI}), evapotranspiration (FD^{MPD}), and human footprint (SR and PD^{MPD}). All diversity-predictor associations and the associated statistics between species groups are provided in [Fig. 3](#), [Supplementary material: Fig. S7](#), and [Supplementary material: Table S4](#).

3.2. Diversity-predictor associations between ecoregions

The diversity-predictor associations were variable between ecoregions, with temperature (*f*: 8), precipitation seasonality and human footprint (*f*: 5 each), precipitation seasonality (*f*: 4), vegetation (*f*: 3), evapotranspiration and precipitation (*f*: 2), and terrain ruggedness (*f*: 1) having the highest relative contribution to the explained diversity variances, although there was still no systematic affiliations between

Table 1

The distribution of terrestrial mammals in Kenya. The table summarises the measures of central tendency for uncorrelated diversity indices, with values representing averages and standard deviations separated by '±'. **(PAs)** summarises diversity indices between ecoregions in the PAs dataset, while **(Grids)** summarises the values for the Grids dataset. F: F statistics, Sig: p-value. SR: species richness, endemism: phylogenetic endemism, PD^{MPD}: phylogenetic mean pairwise distance, FD^{MPD}: functional mean pairwise distance, PD^{NTI}: phylogenetic nearest taxon index, FD^{NTI}: functional nearest taxon index. N: number of PAs/grids.

Dataset	Ecoregion	SR	Endemism	PD ^{MPD}	FD ^{MPD}	PD ^{NTI}	FD ^{NTI}	N
(PAs)	<i>southwestern</i>	159.0 ± 11.5	0.0028 ± 0.0014	161.3 ± 0.7	0.4412 ± 0.0112	0.1291 ± 0.2827	0.0387 ± 0.3479	166
	<i>central-to-southern</i>	137.3 ± 19.9	0.0021 ± 0.0007	161.2 ± 0.8	0.4604 ± 0.0142	−0.1219 ± 0.2287	−0.2256 ± 0.2219	166
	<i>southcoast</i>	122.9 ± 8.8	0.0028 ± 0.0003	160.1 ± 0.9	0.4452 ± 0.0065	−0.1892 ± 0.1644	0.5260 ± 0.2544	36
	<i>northern</i>	102.7 ± 10.9	0.0025 ± 0.0010	160.3 ± 2.1	0.4784 ± 0.0104	0.1874 ± 0.3055	0.2361 ± 0.2393	17
	<i>northcoast</i>	102.4 ± 9.8	0.0030 ± 0.0019	159.1 ± 1.1	0.4833 ± 0.0091	0.1525 ± 0.3428	0.0839 ± 0.3821	14
	F	102.356	6.659	27.569	85.747	22.756	46.564	
	Sig.	0.000	0.000	0.000	0.000	0.000	0.000	
(Grids)	<i>southwestern</i>	156.0 ± 13.5	0.0008 ± 0.0005	161.25 ± 0.83	0.4485 ± 0.0150	0.0566 ± 0.1580	−0.0596 ± 0.1915	595
	<i>northwestern</i>	107.1 ± 14.9	0.0004 ± 0.0001	159.49 ± 1.49	0.4647 ± 0.0173	0.1005 ± 0.1257	−0.0159 ± 0.1426	595
	<i>tanariverbasin</i>	102.8 ± 14.7	0.0004 ± 0.0002	160.1 ± 1.14	0.4800 ± 0.0105	−0.1649 ± 0.1085	0.5260 ± 0.1222	246
	<i>northeastern</i>	63.6 ± 10.5	0.0002 ± 0.0001	160.38 ± 0.83	0.5116 ± 0.0149	0.0366 ± 0.1302	−0.0193 ± 0.1096	562
	<i>coastal</i>	114.4 ± 19.7	0.0007 ± 0.0003	159.48 ± 1.16	0.4661 ± 0.0154	−0.0123 ± 0.1260	0.0667 ± 0.1517	137
	F	3261.305	314.573	206.398	1349.384	176.17	30.968	
	Sig.	0.000	0.000	0.000	0.000	0.000	0.000	

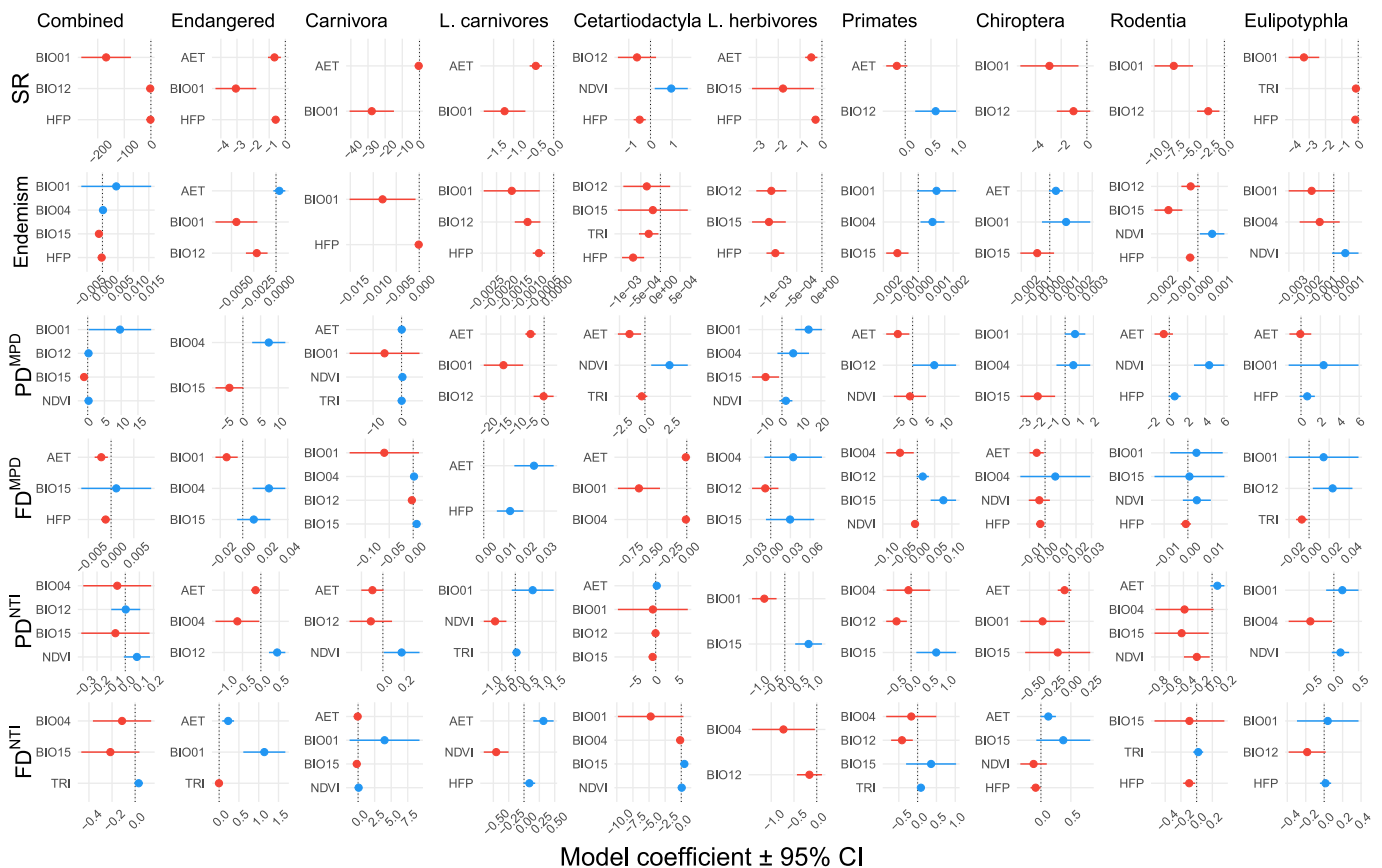


Fig. 3. Associations between diversity variances and abiotic predictors between species groups in protected areas (PAs) in Kenya. The figure illustrates the coefficients of generalized linear mixed models summarising associations, with shape wings representing the 95th percent confidence interval (lower–upper) of model coefficients. Color denotes model effects, with red indicating negative effect and blue indicating positive effect. The predictor variables include; actual annual evapotranspiration (AET), Annual Mean Temperature (BIO01), Annual Precipitation (BIO12), normalized difference vegetation index (NDVI), Precipitation Seasonality (BIO15), Temperature Seasonality (BIO04), terrain (ELEV), and Terrain Ruggedness Index (TRI). (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

predictors and ecoregions (Supplementary material: Fig. S6; Fig. 4; Supplementary material: Table S4).

In the PAs dataset, predictors explained lower variances in *northcoast* ($\bar{x}_{Adj}R^2$: 0.22), followed by *northern* ($\bar{x}_{Adj}R^2$: 0.621), *southwestern* ($\bar{x}_{Adj}R^2$: 0.678), *southcoast* ($\bar{x}_{Adj}R^2$: 0.693), and *central-to-southern* ($\bar{x}_{Adj}R^2$: 0.751). Diversity variances in *southwestern*, were most strongly

associated with temperature, which had the strongest association with all indices except endemism which was most strongly associated with human footprint. In *central-to-southern*, diversity variances had the strongest associations with temperature (SR), precipitation (PD^{NTI}), evapotranspiration (FD^{MPD}), human footprint (PD^{MPD} and FD^{NTI}), and terrain ruggedness (endemism). In *southcoast*, diversity variances had

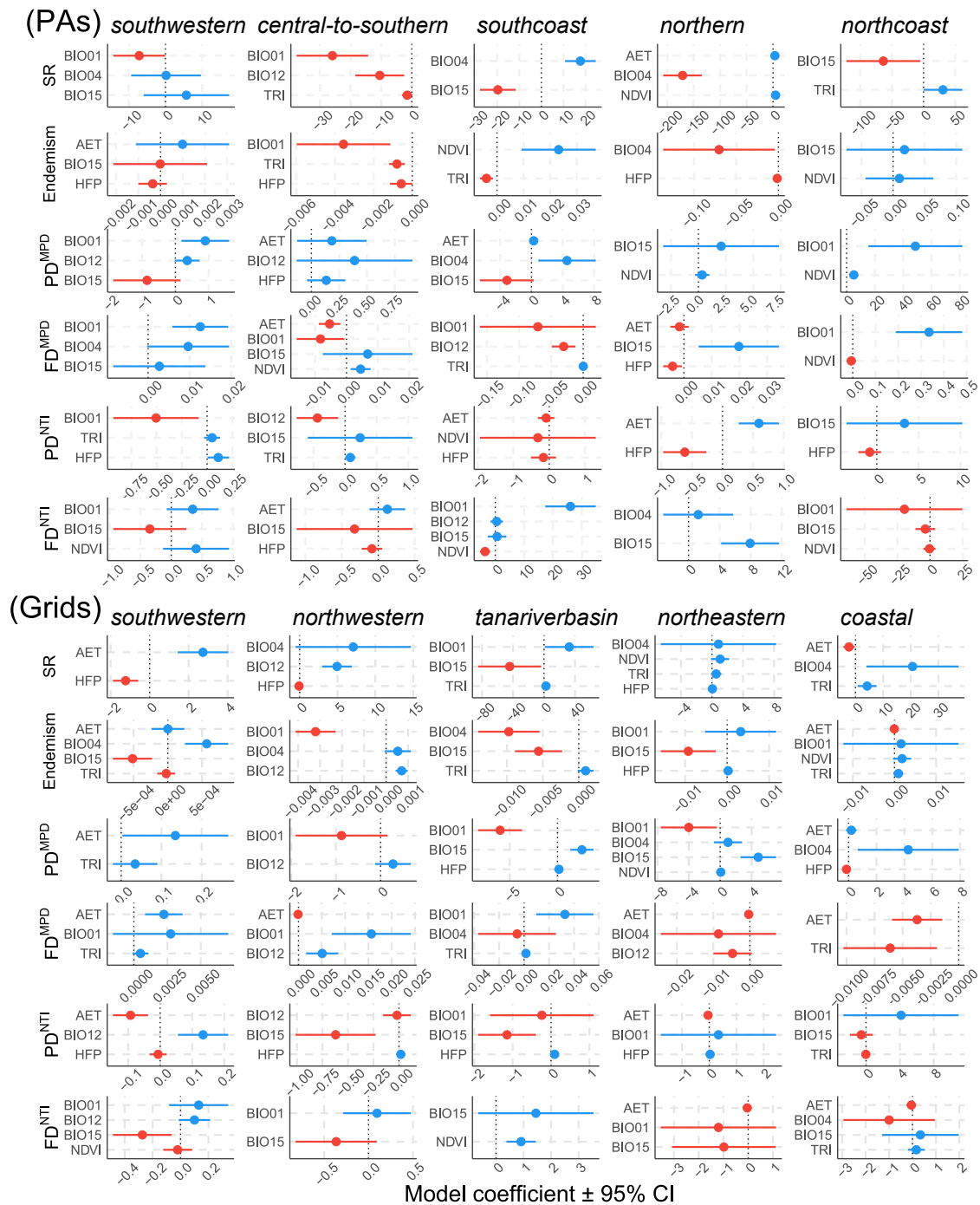


Fig. 4. Associations between diversity variances and abiotic predictors across ecoregions in Kenya. The estimated model coefficients are shown on the x-axis and are represented by points, with bars indicating the lower and upper confidence intervals (95%). Blue-colored points and bars are variables with positive effects on the responses (diversity indices), while red-colored points and bars are variables with negative effects on the responses. The predictor variables include actual annual evapotranspiration (AET), annual mean temperature (BIO01), annual precipitation (BIO12), normalized difference vegetation index (NDVI), precipitation seasonality (BIO15), temperature seasonality (BIO04), terrain (ELEV), and terrain ruggedness index (TRI). (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

the strongest associations with temperature (FD^{NTI}), temperature seasonality (SR and PD^{MPD}), precipitation (FD^{MPD}), vegetation (endemism), and human footprint (PD^{NTI}), which varied slightly with *northcoast*, where temperature (FD^{MPD}), precipitation seasonality (SR, endemism, and FD^{NTI}), vegetation (PD^{MPD}), and human footprint (PD^{NTI}) were the most important predictors. In *northern*, temperature seasonality (SR and endemism), precipitation seasonality (FD^{MPD} and FD^{NTI}), vegetation (PD^{MPD}), and evapotranspiration (PD^{NTI}) had the highest contribution to explained diversity variances (Fig. 4; Supplementary material:

Table S4).

These differing associations between diversity indices and the corresponding strongest predictors persisted in the grid-based analyses, where the explained diversity variances were higher in *northwestern* ($\bar{x}_{Adj}R^2$: 0.854), followed by *southwestern* ($\bar{x}_{Adj}R^2$: 0.852), *northeastern* ($\bar{x}_{Adj}R^2$: 0.787), *coastal* ($\bar{x}_{Adj}R^2$: 0.753), and *tanariverbasin* ($\bar{x}_{Adj}R^2$: 0.684). The strongest predictors included precipitation seasonality (f: 7), temperature and evapotranspiration (f: 6), temperature seasonality (f: 4), precipitation and vegetation (f: 3), and terrain ruggedness (f: 1)

(Fig. 4; Supplementary material: Table S4).

3.3. Human footprint effects on diversity variations

Human footprint was generally negatively associated with diversity indices such that areas of high HFP corresponded to low SR and endemism and high phylogenetic and functional distances between species (Fig. 5; Supplementary material: Fig. S8A-D; Supplementary material: Table S5). While the intermediate disturbance hypothesis was supported in the combined species pool, the various species groups portrayed contrasting associations, mainly hump-shaped or decreasing (Supplementary material: Fig. S8A-D). However, more statistically nonsignificant than significant associations existed between species groups and ecoregions across diversity indices (Supplementary material: Table S5). The strongest effect in the PAs dataset was among large carnivores, combined species pool, Cetartiodactyla, endangered species, and large herbivores, while in the grids dataset, the strongest effect was on Eulipotyphla, Chiroptera, Rodentia, and the combined species pool (Fig. 5; Supplementary material: Table S5). Similarly, the PAs-based analyses between ecoregions showed dominantly negative human footprint effects on species richness but variable effects on the phylogenetic and functional indices (Fig. 5). The strongest effects are in *southwestern*, followed by *central-to-southern*, *southcoast*, *northern*, and weakest in *northcoast*. In the grids-based analyses, the human footprint effect was strongest in *southwestern*, followed by the *northeastern*, *coastal*, *northwestern*, and lowest in *tanariverbasin* (Fig. 5; Supplementary material: Table S5).

4. Discussion

4.1. Unique biotic and anthropogenic influences on terrestrial mammal distribution, diversity, and regionalization patterns in Kenya

Geographical variations in species composition and phylogenetic dissimilarity clustered PAs and grids into distinct ecoregions, notably corresponding to Kenya's unique eco-climatic gradient — hot, humid coastal forests, moist Afromontane forests, and xeric grasslands, bushlands, and thickets. The southwestern diagonal of the country was richer (for taxonomic, phylogenetic, and functional richness) than the northeastern, with the phylogenetic and functional community structure indices revealing rather nonrandom clustering and overdispersion tendencies across the country. The temperature and precipitation (annual mean and seasonality) and human footprint emerged as consistently strongly correlated with diversity variances, suggesting they were the most important predictors of terrestrial mammal diversity patterns in Kenya. The notable diversity-predictor variations between species groups, ecoregions, and diversity indices, broadly concur with substantiated accounts of strong associations between mammalian assembling patterns in the tropics to ecological-evolutionary conservatism and climatic sustenance of habitat productivity (Brown, 2014; Rowan et al., 2020), and are comparable to previous findings on vertebrate diversity associations with climatic, environmental, and human variables (Kissling et al., 2012; Montaña-Centellas et al., 2020; Montaña-Centellas et al., 2022; Onditi et al., 2022; Ramesh et al., 2016; Rowe, 2009). Species trait flexibility to ecological conditions is one of the fundamental factors underlining association variabilities between diversity patterns and key abiotic determinants across species groups and ecoregions, where traits such as diet and activity patterns are tuned in response not only to competitive interactions but also to environmental constraints and opportunities. In relation, we observed distinct ecoregions formed along eco-climatic (topography and climate), reinforcing a dominant role of climate and orogeny in the ecological regionalization of faunal communities in Kenya. The ecoregions' formations resemble phyto-geographical regions (Zhang et al., 2022), implying intimate associations between plant and mammal diversity patterns, resource needs, and prey-predator dynamics. The observed diversity-HFP associations agree

with recent observations of increasingly patterned detrimental influences of human-wildlife interactions underlying human-induced defaunation (Dirzo et al., 2014; Wen et al., 2020). Such influence on species imperilment can be more substantial than environmental factors (Dirzo et al., 2014; Venter et al., 2016), just as our findings also revealed. Apparently, efforts to curb the direct impacts of human activities that directly extract wildlife or degrade vital ecosystems to endanger their survival should be ingrained into national biodiversity management plans to protect terrestrial mammals adequately.

4.2. Variable diversity-predictor associations between species groups and ecoregions

Variable diversity-predictor associations between species groups and ecoregions characterized Kenya's terrestrial mammal diversity patterns. For one, the contrasts between species groups suggest that the combined species pool may not accurately surrogate diversity-predictor associations for specific species clades and guilds groups, especially when geographical regionalization is considered (Ding et al., 2019; Gray et al., 2007). Similarly, Andrews and O'Brien (2000) observed patterns where only the SR of large mammals had a strong correlation with annual temperature compared to the combined mammal species pool in southern Africa, further supporting that guild- and clade-specific constraints in ecological and evolutionary opportunities are strong determinants of terrestrial mammal diversity patterns (Ding et al., 2019), and Kissling et al. (2012) found substantial variations in species richness patterns across guilds based on geographical and environmental context, highlighting the marked influence of ecological and evolutionary opportunities and constraints. The distribution disparities between diversity indices and differences in strongest predictors suggest species richness is poorly interchangeable for interpretations of evolutionary and trait diversity dynamics, reiterating the need to understand the particular associations (Brun et al., 2019). Also, these patterns highlight systematic diversity-predictor associations, where the phylogenetic and functional diversity patterns are not represented adequately by species richness, thus, must be intentionally targeted in biodiversity conservation prioritization (Mazel et al., 2018; Oberosler et al., 2020; Oliveira et al., 2016; Onditi et al., 2021; Onditi et al., 2022).

4.3. Water-energy dynamics and human footprint are the most important predictors of terrestrial mammal diversity patterns in Kenya

In the tropics, historical and long-term invariability in climatic regimes supported long-term, stably productive habitats, which strongly determined vertebrate assembling, but with relatively weaker environmental filtering effects overall (Fritz et al., 2016; Martínez-Meyer et al., 2004). Here, we observed the bioclimatic variables (precipitation and temperature annual averages and seasonality) systematically retained as most strongly associated with the various diversity indices. Such systematically strong associations between mammal diversity and climate broadly support the water-energy dynamics hypothesis as a good explainer of terrestrial mammal diversity patterns in Kenya. Previous studies in tropical systems also recorded similar associations between water availability and temperature with vertebrate biodiversity patterns (Brown, 2014; Hawkins et al., 2003). Our findings extend the understanding that precipitation and temperature regimes have strong effects on terrestrial mammal diversity patterns even at relatively local spatial scales such as the extent of Kenya (and by inference, the tropical-equatorial geographical context), and underlie the high mammal biodiversity of the Afrotropic (Fordham et al., 2019; Theodoridis et al., 2020). Notably, the vegetation and terrain ruggedness variables accounted for low proportions of explained diversity variances across species groups and ecoregions, implying that the habitat heterogeneity hypothesis might be a poor fit in explaining the terrestrial mammal diversity patterns in Kenya. Habitat productivity has been observed to be highly correlated to patterns in vertebrate distribution and assembling,

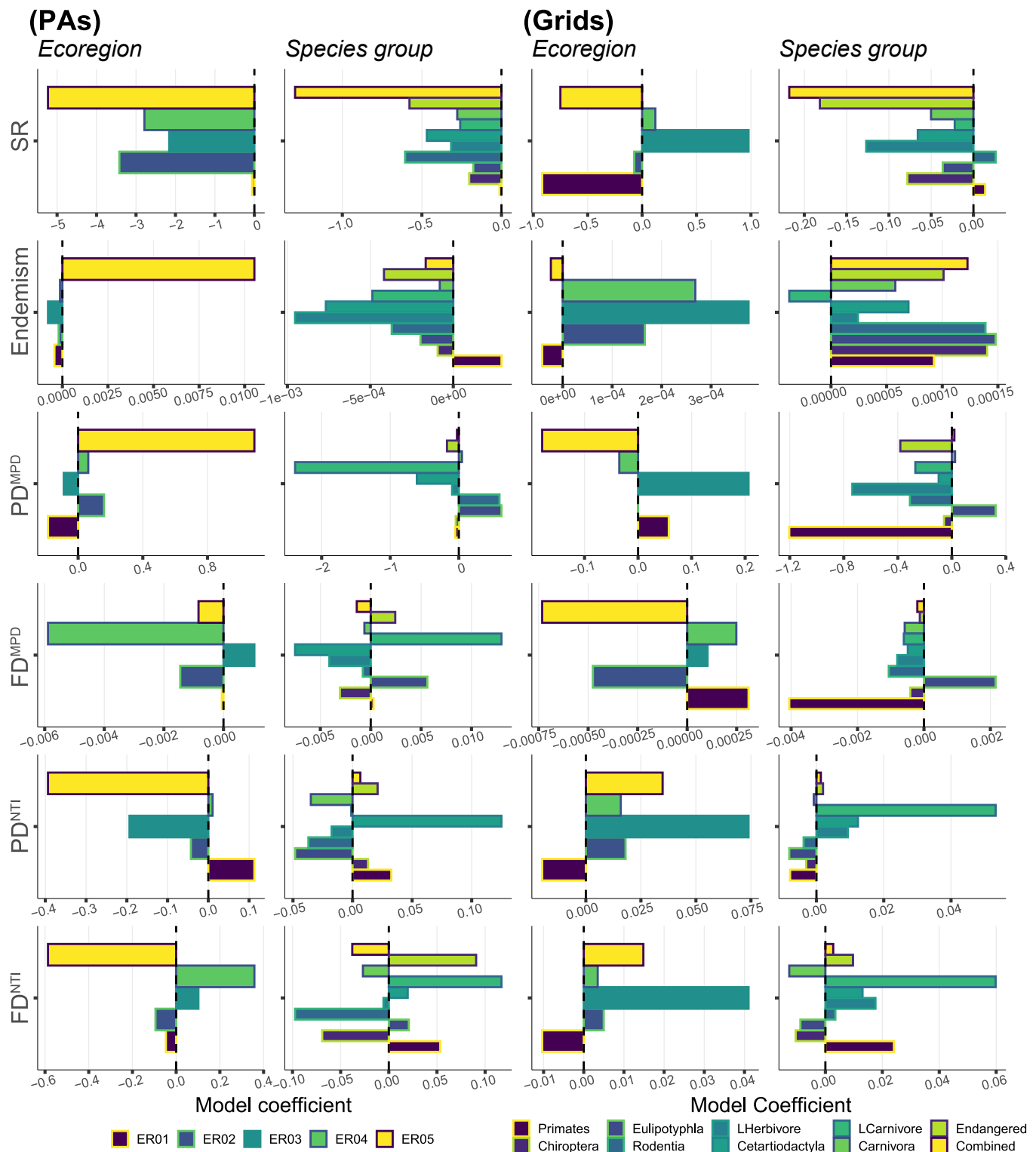


Fig. 5. Association of human footprint to terrestrial mammal diversity variances. Top figure: between ecoregions. PAs show the PAs-based, while Grids show the grid-based analysis. See [Supplementary material: Fig. S8A-C](#) for the smoothed scatter (XY) plots for human footprint relationships with diversity indices across species groups and ecoregions. The relative proportion of diversity indices accounted for by human footprint when combined with other predictors and retained in the best-fit models are shown in [Fig. 4](#) and [Fig. 5](#). In (PAs) the legend abbreviations include; ER01 = southwestern, ER02 = central-to-southern, ER03 = southcoast, ER04 = northern, and ER05 = northcoast, while in (Grids) they match; ER01 = southwestern, ER02 = northwestern, ER03 = tanariverbasin, ER04 = northeastern, and ER05 = coastal.

but only when direct water-energy variables are controlled for or excluded from the analyses (Jetz et al., 2009; McCain et al., 2018; Qian, 2009). Furthermore, similar responses of plant and animal evolution and distribution to water and energy availability might confound functional ecological linkages between them. Despite terrain ruggedness having a relatively weak correlation with explained diversity variances across species groups and ecoregions, topography inherently influences local abiotic and biotic interactions within habitats and plays a vital role in species' diversity, distributions, and historical evolutionary radiation (Badgley, 2010; Stein et al., 2014). Kenya's topography's evolutionary and ecological implication on mammal diversity patterns is notable (Demos et al., 2014; Jones and Safi, 2011; Onditi et al., 2022), often illustrated by the highest faunal and floral diversity concentrated in the relatively topographically complex central and western Kenya regions, where the landscape is dominated by high peaks (Mount Kenya, Aberdare Ranges, Mount Elgon, and Cherangani Hills) and highlands (Loita Hills and Mau Forests).

The variations in the strength of diversity-predictor associations between ecoregions appear systematic to the filtering influence of abiotic vs. biotic interactions across environmental gradients. For one, while the climatic conditions in the northern and north-eastern regions are characterized by high temperatures, low annual precipitation, and higher seasonality in temperature and precipitation, making them *harsh/stressful* for most mammals, the central to western regions are characterized by high precipitation, more vegetation, and high terrain ruggedness, making them relatively *benign* for mammal persistence. Under natural conditions, harsh habitats strongly select species ecologically well adapted to the environmental thresholds, leading to assemblages that are more phylogenetically and functionally clustered as species converge traits to similar ecological functions. In benign habitats, on the other hand, high resource availability implies weak environmental filtering, and species coexistence dynamics drive unique ecological functions through trait dispersion (competitive interactions and niche specialization), leading to assemblages that are more phylogenetically and functionally more unique/overdispersed (Cooke et al., 2019; Gorczynski et al., 2021; Le Bagousse-Pinguet et al., 2019; Oliveira et al., 2016; Safi et al., 2011). These ecological adaptations for coexistence across habitat suitability gradients align with the observed high species richness versus the phylogenetic and functional community structure in the more heterogeneous (more productive) ecoregions/habitats.

We found a dominantly detrimental effect of human activity pressure on terrestrial mammal diversity, where species richness and endemism decreased, and mean phylogenetic and functional pairwise distances increased as human footprint increased. As such, communities in areas with high human footprint had fewer species with relatively distant phylogenetic and functional affiliations, typical of impoverished ecosystems, findings which reiterate the harmful interactions between human activities and terrestrial mammal diversity (Ogutu et al., 2016; Veldhuis et al., 2019). The richness-dependent indices were more strongly influenced than the richness-controlled ones, suggesting variable species tolerance to human activity pressure based on ecosystem functions and evolutionary history (Andrews and O'Brien, 2000). Despite the intermediate disturbance hypothesis being supported in the combined species pool, contrasting associations emerged between species groups and ecoregions. These findings add evidence of the destructive impacts of human activity on wildlife (Fabricius et al., 2003; Gray et al., 2016; Rada et al., 2019; Veldhuis et al., 2019; Li et al., 2021). Like several Sub-Saharan countries, Kenya's fast-paced human population growth and urbanization drive encroachment into previously relatively pristine wildlife areas for agriculture, settlement, and infrastructural expansion land-use conversions (Lindsey et al., 2022). From an ecological perspective, such intensifying human footprint on wildlife areas progressively constrains the natural regeneration of habitats by pushing wildlife to restrictive core areas; this threatens the ecosystems' natural ecological sustainability by compromising the

natural resource regeneration rates and leads to the wildlife emigrating to human-dominated areas (Mukeka et al., 2019; Ogutu et al., 2016; Veldhuis et al., 2019). Also, some PAs, such as the Nairobi National Park, are surrounded by intensively built-up areas, while others with high tourism potential, such as the Masai Mara National Reserve, attract high infrastructural developments around them, which reciprocally attracts human activities. Such inherent proximity of human settlement areas and priority wildlife areas such as PAs might confound the effect of human activity pressure on terrestrial mammal distribution and diversity patterns in studies such as this.

5. Study limitations

Our findings must be interpreted within the potentials and constraints of the input data and analyses. For one, we used species ranges and occurrence records from IUCN and GBIF as primary data on the mammal fauna composition in Kenya for lack of high-resolution nationwide distribution data. Despite the precautions taken to ensure that the species ranges matched the latest national taxonomic and geographic accounts of mammals of Kenya, such as Musila et al. (2019), these representations might still be flawed appraisals of species-specific habitat usages. Nonetheless, they provide an essential assessment of biodiversity distributional patterns and are a vital baseline for spatial conservation prioritization (Chape et al., 2005). Another potential confounding factor is the spatial resolution of the raw input data. Our pre-analysis data preparation ensured that the resolution and coverage were highly representative of local conditions and comparable to currently available national-level climate data (such as the high-resolution ensemble climate models for Africa (Philip et al., 2015) and the national ecoclimatic characterizations (Western et al., 2015)). Thus, our analyses of abiotic factors driving terrestrial mammal diversity are relatively reliable. Also, the global human footprint data used in our study (Venter et al., 2016; Venter et al., 2018) are currently the best publicly available nationwide mapping of human impacts in wildlife areas despite human activity influences on terrestrial mammal diversity likely not adequately captured by such an index since livestock grazing, firewood logging, and collecting bio-resources occur at smaller local scales; the data probably underestimates local-scale human influence on wildlife diversity and activity patterns (Harfoot et al., 2021). However, recent comparative tests have shown that the data perform better than currently available related datasets (Kennedy et al., 2019; Venter et al., 2020).

5.1. Conclusion

We investigated the main abiotic predictors (environmental and anthropogenic) of multiple diversity dimensions of terrestrial mammals in Kenya and how they vary geographically and between species groups (guilds and clades). The temperature and precipitation variables and human footprint emerged as consistently highly correlated with diversity variances, consistent with previous studies, although associations between diversity indices and the respective strongest predictors were not systematic between species groups and ecoregions, suggesting the interactions are more complex than currently understood. These findings highlight a scenario where the interplay of climate and human footprint influences systematically determines multiple dimensions of terrestrial mammal diversity. Thus, for adequate sustained protection of terrestrial mammals, efforts to curb human activities that degrade climatic regimes should be entrenched in biodiversity management programs. Species richness and its strongest predictors were poor surrogates for the phylogenetic and functional diversity dimensions. Thus, phylogenetic and trait diversity must be deliberately targeted by conservation efforts to assuage the preservation of the integrity of ecosystems. Our findings extend the understanding of the abiotic predictors of terrestrial mammal diversity to the national scale, where most ecosystem and terrestrial mammal management policies and actions are most

applicable; these are highly relevant to Kenya's nationally-tailored biodiversity management systems. The effectiveness of various biodiversity conservation initiatives can significantly benefit from drawing action plans that meaningfully consider species' ecological dynamics between clades and guilds.

CRedit authorship contribution statement

Kenneth Otieno Onditi: Conceptualization, Methodology, Formal analysis, Investigation, Resources, Data curation, Visualization, Writing – original draft, Writing – review & editing. **Wenyu Song:** Conceptualization, Resources, Data curation, Writing – review & editing. **Xueyou Li:** Conceptualization, Validation, Resources, Writing – review & editing. **Simon Musila:** Investigation, Resources, Writing – review & editing. **Zhongzheng Chen:** Investigation, Writing – review & editing. **Quan Li:** Investigation, Writing – review & editing. **James Mathenge:** Investigation, Resources, Writing – review & editing. **Esther Kioko:** Investigation, Resources, Writing – review & editing, Supervision. **Xuelong Jiang:** Conceptualization, Validation, Resources, Writing – review & editing, Supervision.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

All data is contained within the article and its supplementary material. These are also available in FigShare (<https://doi.org/10.6084/m9.figshare.16573418>), including the R codes.

Acknowledgements

The study is associated with mammal surveys conducted in Kenya between 2015 and 2019 by joint research teams from the National Museums of Kenya (NMK) and Kunming Institute of Zoology, Chinese Academy of Sciences (KIZ). We are particularly grateful to the personnel of the Mammalogy section at NMK and the KIZ Mammal Ecology and Evolution Research Group for facilitating logistical support during fieldwork.

Funding

The mammal survey fieldwork in Kenya associated with the study was funded by the Sino-Africa Joint Research Centre, Chinese Academy of Sciences (SAJC202103), which also funded the APC. The CAS-TWAS President's Fellowship Program sponsored KOO.

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.ecolind.2023.110595>.

References

- Allouche, O., Kalyuzhny, M., Moreno-Rueda, G., Pizarro, M., Kadmon, R., 2012. Area-heterogeneity tradeoff and the diversity of ecological communities. *Proc. Natl. Acad. Sci. U.S.A.* 109 (43), 17495–17500. <https://doi.org/10.1073/pnas.1208652109>.
- Amatulli, G., Domisch, S., Tuanmu, M.N., Parmentier, B., Ranipeta, A., Malczyk, J., Jetz, W., 2018. A suite of global, cross-scale topographic variables for environmental and biodiversity modeling. *Sci. Data* 5, 180040. <https://doi.org/10.1038/sdata.2018.40>.
- Andermann, T., Faurby, S., Turvey, S.T., Antonelli, A., Silvestro, D., 2020. The past and future human impact on mammalian diversity. *Sci. Adv.* 6, eabb2313 <https://doi.org/10.1126/sciadv.abb2313>.
- Andrews, P., O'Brien, E.M., 2000. Climate, vegetation, and predictable gradients in mammal species richness in southern Africa. *J. Zool.* 251, 205–231. <https://doi.org/10.1111/j.1469-7998.2000.tb00605.x>.
- Badgley, C., 2010. Tectonics, topography, and mammalian diversity. *Ecography* 33, 220–231. <https://doi.org/10.1111/j.1600-0587.2010.06282.x>.
- Barrett, C.B., Travis, A.J., Dasgupta, P., 2011. On biodiversity conservation and poverty traps. *Proc. Natl. Acad. Sci. U.S.A.* 108, 13907–13912. <https://doi.org/10.1073/pnas.1011521108>.
- Barton, K., 2019. MuMIn: Multi-Model Inference. R package version 1.43.6., Model selection and model averaging based on information criteria. Retrieved from. <http://CRAN.R-project.org/package=MuMIn>.
- Brown, J.H., 2014. Why are there so many species in the tropics? *J. Biogeogr.* 41 (1), 8–22. <https://doi.org/10.1111/jbi.12228>.
- Brun, P., Zimmermann, N.E., Graham, C.H., Lavergne, S., Pellissier, L., Münckmüller, T., Thuiller, W., 2019. The productivity-biodiversity relationship varies across diversity dimensions. *Nat. Commun.* 10, 5691. <https://doi.org/10.1038/s41467-019-13678-1>.
- Burgin, C.J., Wilson, D.E., Mittermeier, R.A., Rylands, A.B., Lacher, T.E., Sechrest, W., 2020. Illustrated checklist of the mammals of the world. Lynx Edicions, Barcelona, Spain.
- Burnham, K.P., Anderson, D.R. (Eds.), 2004. *Model Selection and MultiModel Inference*. Springer New York, New York, NY.
- Calcagno, V., 2019. glmulti: Model Selection and Multimodel Inference Made Easy. R package version 1.0.7.1. Retrieved from <https://CRAN.R-project.org/package=glmulti>.
- Cazalis, V., 2022. Species richness response to human pressure hides important assemblage transformations. *Proc. Natl. Acad. Sci. U.S.A.* 119 (19), e2107361119 <https://doi.org/10.1073/pnas.2107361119>.
- Ceballos, G., Ehrlich, P.R., Barnosky, A.D., Garcia, A., Pringle, R.M., Palmer, T.M., 2015. Accelerated modern human-induced species losses: Entering the sixth mass extinction. *Sci. Adv.* 1, e1400253 <https://doi.org/10.1126/sciadv.1400253>.
- Chape, S., Harrison, J., Spalding, M., Lysenko, I., 2005. Measuring the extent and effectiveness of protected areas as an indicator for meeting global biodiversity targets. *Philos. Trans. R. Soc., B* 360, 443–455. <https://doi.org/10.1098/rstb.2004.1592>.
- Connell, J.H., 1978. Diversity in Tropical Rain Forests and Coral Reefs. *Science* 199 (4335), 1302–1310. <https://doi.org/10.1126/science.199.4335.1302>.
- Cooke, R.S.C., Bates, A.E., Eigenbrod, F., Schrodt, F., 2019. Global trade-offs of functional redundancy and functional dispersion for birds and mammals. *Glob. Ecol. Biogeogr.* 28, 484–495. <https://doi.org/10.1111/geb.12869>.
- Cornwell, W.K., Schwill, L.D., Ackerly, D.D., 2006. A trait-based test for habitat filtering: convex hull volume. *Ecology* 87, 1465–1471. [https://doi.org/10.1890/0012-9658\(2006\)87\[1465:atfthf\]2.0.co;2](https://doi.org/10.1890/0012-9658(2006)87[1465:atfthf]2.0.co;2).
- Couvreur, T.L.P., Dauby, G., Blach-Overgaard, A., Deblauwe, V., Dessein, S., Droissart, V., Hardy, O.J., Harris, D.J., Janssens, S.B., Ley, A.C., Mackinder, B.A., Sonké, B., Sosef, M.S.M., Stévant, T., Svenning, J.-C., Wieringa, J.J., Faye, A., Missoup, A.D., Tolley, K.A., Nicolas, V., Ntse, S., Fluteau, F., Robin, C., Guillocheau, F., Barboni, D., Sepulchre, P., 2021. Tectonics, climate and the diversification of the tropical African terrestrial flora and fauna. *Biol. Rev. Camb. Philos. Soc.* 96, 16–51. <https://doi.org/10.1111/brv.12644>.
- Daru, B.H., Karunaratne, P., Schliep, K., Silvestro, D., 2020. phyloregion: R package for biogeographical regionalization and macroecology. *Methods Ecol. Evol.* 11 (11), 1483–1491. <https://doi.org/10.1111/2041-210X.13478>.
- Demos, T.C., Kerbis Peterhans, J.C., Agwanda, B., Hickerson, M.J., 2014. Uncovering cryptic diversity and refugial persistence among small mammal lineages across the Eastern Afrotropical biodiversity hotspot. *Mol. Phylogenet. Evol.* 71, 41–54. <https://doi.org/10.1016/j.ympev.2013.10.014>.
- Didan, K., 2015. MOD13A3 MODIS/Terra vegetation Indices Monthly L3 Global 1km SIN Grid V006. NASA EOSDIS Land Processes DAAC 10. <https://doi.org/10.5067/MODIS/MOD13A3.006>.
- Ding, Z., Liang, J., Hu, Y., Zhou, Z., Sun, H., Liu, L., Liu, H., Hu, H., Si, X., 2019. Different responses of avian feeding guilds to spatial and environmental factors across an elevation gradient in the central Himalaya. *Ecol. Evol.* 9, 4116–4128. <https://doi.org/10.1002/ece3.5040>.
- Dirzo, R., Young, H.S., Galetti, M., Ceballos, G., Isaac, N.J., Collen, B., 2014. Defaunation in the Anthropocene. *Science* 345, 401–406. <https://doi.org/10.1126/science.1251817>.
- Dormann, C.F., 2007. Effects of incorporating spatial autocorrelation into the analysis of species distribution data. *Glob. Ecol. Biogeogr.* 16, 129–138. <https://doi.org/10.1111/j.1466-8238.2006.00279.x>.
- Dormann, C.F., Calabrese, J.M., Guillerá-Aroita, G., Matechou, E., Bahn, V., Bartoň, K., Beale, C.M., Ciuti, S., Elith, J., Gerstner, K., Guelat, J., Keil, P., Lahoz-Monfort, J.J., Pollock, L.J., Reineking, B., Roberts, D.R., Schröder, B., Thuiller, W., Warton, D.I., Wintle, B.A., Wood, S.N., Wüest, R.O., Hartig, F., 2018. Model averaging in ecology: a review of Bayesian, information-theoretic, and tactical approaches for predictive inference. *Ecol. Monogr.* 88, 485–504. <https://doi.org/10.1002/ecm.1309>.
- Erwin, D.H., 2009. Climate as a driver of evolutionary change. *Curr. Biol.* 19, R575–R583. <https://doi.org/10.1016/j.cub.2009.05.047>.
- Fabricius, C., Burger, M., Hockey, P.A.R., 2003. Comparing biodiversity between protected areas and adjacent rangeland in xeric succulent thickets, South Africa: arthropods and reptiles. *J. Appl. Ecol.* 40, 392–403. <https://doi.org/10.1046/j.1365-2664.2003.00793.x>.
- Faith, D.P., 1992. Conservation evaluation and phylogenetic diversity. *Biol. Conserv.* 61, 1–10. [https://doi.org/10.1016/0006-3207\(92\)91201-3](https://doi.org/10.1016/0006-3207(92)91201-3).

- Ferrer-Castán, D., Morales-Barbero, J., Vetaas, O.R., 2016. Water-energy dynamics, habitat heterogeneity, history, and broad-scale patterns of mammal diversity. *Acta Oecol.* 77, 176–186. <https://doi.org/10.1016/j.actao.2016.10.010>.
- Fine, P.V.A., 2015. Ecological and evolutionary drivers of geographic variation in species diversity. *Annu. Rev. Ecol. Syst.* 46, 369–392. <https://doi.org/10.1146/annurev-ecolsys-112414-054102>.
- Fordham, D.A., Brown, S.C., Wigley, T.M.L., Rahbek, C., 2019. Cradles of diversity are unlikely relics of regional climate stability. *Curr. Biol.* 29, R356–R357. <https://doi.org/10.1016/j.cub.2019.04.001>.
- Fritz, S.A., Eronen, J.T., Schnitzler, J., Hof, C., Janis, C.M., Mulch, A., Böhning-Gaese, K., Graham, C.H., 2016. Twenty-million-year relationship between mammalian diversity and primary productivity. *Proc. Natl. Acad. Sci. U.S.A.* 113 (39), 10908–10913. <https://doi.org/10.1073/pnas.1602145113>.
- GBIF.org, 2020. Global Biodiversity Information Facility (GBIF) Occurrence Download. Retrieved from <https://doi.org/10.15468/dl.gblnbn>.
- Gillman, L.N., Wright, S.D., 2014. Species richness and evolutionary speed: the influence of temperature, water and area. *J. Biogeogr.* 41 (1), 39–51. <https://doi.org/10.1111/jbi.12173>.
- Gorczynski, D., Hsieh, C., Luciano, J.T., Ahumada, J., Espinosa, S., Johnson, S., Rovero, F., Santos, F., Andrianarisoa, M.H., Astaiza, J.H., Jansen, P.A., Kayijamahe, C., Moreira Lima, M.G., Salvador, J., Beaudrot, L., 2021. Tropical mammal functional diversity increases with productivity but decreases with anthropogenic disturbance. *Proc. R. Soc. B Biol. Sci.* 288, 20202098 <https://doi.org/10.1098/rspb.2020.2098>.
- Gotelli, N.J., Entsminger, G.L., 2003. Swap algorithms in null model analysis. *Ecology* 84, 532–535. [https://doi.org/10.1890/0012-9658\(2003\)084\[0532:Sainma\]2.0.CO;2](https://doi.org/10.1890/0012-9658(2003)084[0532:Sainma]2.0.CO;2).
- Gray, M.A., Baldauf, S.L., Mayhew, P.J., Hill, J.K., 2007. The response of avian feeding guilds to tropical forest disturbance. *Conserv. Biol.* 21, 133–141. <https://doi.org/10.1111/j.1523-1739.2006.00557.x>.
- Gray, C.L., Hill, S.L., Newbold, T., Hudson, L.N., Borger, L., Contu, S., Hoskins, A.J., Ferrier, S., Purvis, A., Scharlemann, J.P., 2016. Local biodiversity is higher inside than outside terrestrial protected areas worldwide. *Nat. Commun.* 7, 12306. <https://doi.org/10.1038/ncomms12306>.
- Hallett, T.B., Coulson, T., Pilkington, J.G., Clutton-Brock, T.H., Pemberton, J.M., Grenfell, B.T., 2004. Why large-scale climate indices seem to predict ecological processes better than local weather. *Nature* 430, 71–75. <https://doi.org/10.1038/nature02708>.
- Harfoot, M.B.J., Johnston, A., Balmford, A., Burgess, N.D., Butchart, S.H.M., Dias, M.P., Hazin, C., Hilton-Taylor, C., Hoffmann, M., Isaac, N.J.B., Iversen, L.L., Outhwaite, C. L., Visconti, P., Geldmann, J., 2021. Using the IUCN Red List to map threats to terrestrial vertebrates at global scale. *Nat. Ecol. Evol.* 5 (11), 1510–1519. <https://doi.org/10.1038/s41559-021-01542-9>.
- Hawkins, B.A., Field, R., Cornell, H.V., Currie, D.J., Guegan, J.F., Kaufman, D.M., Kerr, J. T., Mittelbach, G.G., Oberdorff, T., O'Brien, E.M., Porter, E.E., Turner, J.R.G., 2003. Energy, water, and broad-scale geographic patterns of species richness. *Ecology* 84, 3105–3117. <https://doi.org/10.1890/03-8006>.
- Jetz, W., Kreft, H., Ceballos, G., Mutke, J., 2009. Global associations between terrestrial producer and vertebrate consumer diversity. *Proc. R. Soc. B.* 276 (1655), 269–278. <https://doi.org/10.1098/rspb.2008.1005>.
- Jones, K.E., Safi, K., 2011. Ecology and evolution of mammalian biodiversity. *Philos. Trans. R. Soc. B* 366, 2451–2461. <https://doi.org/10.1098/rstb.2011.0090>.
- Jones, K.E., Bielby, J., Cardillo, M., Fritz, S.A., O'Dell, J., Orme, C.D.L., Safi, K., Sechrest, W., Boakes, E.H., Carbone, C., Connolly, C., Cutts, M.J., Foster, J.K., Grenyer, R., Habib, M., Plaster, C.A., Price, S.A., Rigby, E.A., Rist, J., Teacher, A., Bininda-Emonds, O.R.P., Gittleman, J.L., Mace, G.M., Purvis, A., Michener, W.K., 2009. PANTHERIA: a species-level database of life history, ecology, and geography of extant and recently extinct mammals. *Ecology* 90, 2648. <https://doi.org/10.1890/08-1494.1>.
- Kang, S., Niu, J., Zhang, Q., Li, D., Ren, H., Ren, J., Zhang, X., Dong, J., 2016. Environmental filtering does not necessarily prevent trait divergence: a case study of the Xilin River Basin in Inner Mongolia, China. *J. Plant Ecol.* 10, 497–509. <https://doi.org/10.1093/jpe/rtw050>.
- Karger, D.N., Conrad, O., Böhner, J., Kawohl, T., Kreft, H., Soria-Auza, R.W., Zimmermann, N.E., Linder, H.P., Kessler, M., 2017. Climatologies at high resolution for the earth's land surface areas. *Sci. Data* 4, 170122. <https://doi.org/10.1038/sdata.2017.122>.
- Kennedy, C.M., Oakleaf, J.R., Theobald, D.M., Baruch-Mordo, S., Kiesecker, J., 2019. Managing the middle: A shift in conservation priorities based on the global human modification gradient. *Glob. Chang. Biol.* 25, 811–826. <https://doi.org/10.1111/gcb.14549>.
- Kerr, J., Packer, L., 1997. Habitat heterogeneity as a determinant of mammal species richness in high-energy regions. *Nature* 385, 252–254. <https://doi.org/10.1038/385252a0>.
- Kingdon, J., Butynski, T.M., Hapold, D.C.D., Hapold, M., Hoffmann, M., 2013. *Mammals of Africa*. Bloomsbury, London, United Kingdom.
- Kissling, W.D., Sekercioglu, C.H., Jetz, W., 2012. Bird dietary guild richness across latitudes, environments and biogeographic regions. *Glob. Ecol. Biogeogr.* 21, 328–340. <https://doi.org/10.1111/j.1466-8238.2011.00679.x>.
- Kumar, S., Stecher, G., Suleski, M., Hedges, S.B., 2017. TimeTree: A resource for timelines, timetrees, and divergence times. *Mol. Biol. Evol.* 34, 1812–1819. <https://doi.org/10.1093/molbev/msx116>.
- Lacher, T.E., Davidson, A.D., Fleming, T.H., Gómez-Ruiz, E.P., McCracken, G.F., Owen-Smith, N., Peres, C.A., Vander Wall, S.B., 2019. The functional roles of mammals in ecosystems. *J. Mammal.* 100, 942–964. <https://doi.org/10.1093/jmammal/gyy183>.
- Le Bagousse-Pinguet, Y., Soliveres, S., Gross, N., Torices, R., Berdugo, M., Maestre, F.T., 2019. Phylogenetic, functional, and taxonomic richness have both positive and negative effects on ecosystem multifunctionality. *Proc. Natl. Acad. Sci. U.S.A.* 116 (17), 8419–8424. <https://doi.org/10.1073/pnas.1815727116>.
- Legendre, P., 1993. Spatial autocorrelation: Trouble or new paradigm? *Ecology* 74, 1659–1673. <https://doi.org/10.2307/1939924>.
- Legendre, P., Anderson, M.J., 1999. Distance-based redundancy analysis: Testing multispecies responses in multifactorial ecological experiments. *Ecol. Monogr.* 69, 1–24. [https://doi.org/10.1890/0012-9615\(1999\)069\[0001:Dbratm\]2.0.CO;2](https://doi.org/10.1890/0012-9615(1999)069[0001:Dbratm]2.0.CO;2).
- Li, X., Hu, W., Bleisch, W.V., Li, Q., Wang, H., Lu, W., Sun, J., Zhang, F., Ti, B., Jiang, X., 2021. Functional diversity loss and change in nocturnal behavior of mammals under anthropogenic disturbance. *Conservation Biology*. <https://doi.org/10.1111/cobi.13839>.
- Li, L., Wang, Z., Zerbe, S., Abdusali, N., Tang, Z., Ma, M., Yin, L., Mohammad, A., Han, W., Fang, J., Chen, H.Y.H., 2013. Species richness patterns and water-energy dynamics in the drylands of Northwest China. *PLoS One* 8 (6), e66450. <https://doi.org/10.1371/journal.pone.0066450>.
- Lindsey, P.A., Anderson, S.H., Dickman, A., Gandiwa, P., Harper, S., Morakinyo, A.B., Nyambe, N., O'Brien-Onyeka, M., Packer, C., Parker, A.H., Robson, A.S., Ruhweza, A., Sogbohossou, E.A., Steiner, K.W., Tumenta, P.N., 2022. Shepherding Sub-Saharan Africa's Wildlife Through Peak Anthropogenic Pressure Toward a Green Anthropocene. *Annu. Rev. Env. Resour.* 47, 91–121. <https://doi.org/10.1146/annurev-environ-120920-125536>.
- Luo, Z., Tang, S., Li, C., Fang, H., Hu, H., Yang, J., Ding, J., Jiang, Z., White, E.P., 2012. Environmental effects on vertebrate species richness: testing the energy, environmental stability and habitat heterogeneity hypotheses. *PLoS One* 7 (4), e35514. <https://doi.org/10.1371/journal.pone.0035514>.
- MacArthur, R., MacArthur, J.W., 1961. On bird species-diversity. *Ecology* 42, 594–1000. <https://doi.org/10.2307/1932254>.
- Malhi, Y., Franklin, J., Seddon, N., Solan, M., Turner, M.G., Field, C.B., Knowlton, N., 2020. Climate change and ecosystems: threats, opportunities and solutions. *Philos. Trans. R. Soc. B* 375, 20190104. <https://doi.org/10.1098/rstb.2019.0104>.
- Martínez-Meyer, E., Townsend Peterson, A., Hargrove, W.W., 2004. Ecological niches as stable distributional constraints on mammal species, with implications for Pleistocene extinctions and climate change projections for biodiversity. *Glob. Ecol. Biogeogr.* 13, 305–314. <https://doi.org/10.1111/j.1466-822X.2004.00107.x>.
- Mazel, F., Pennell, M.W., Cadotte, M.W., Diaz, S., Dalla Riva, G.V., Grenyer, R., Leprieux, F., Mooers, A.O., Mouillot, D., Tucker, C.M., Pearse, W.D., 2018. Prioritizing phylogenetic diversity captures functional diversity unreliably. *Nat. Commun.* 9, 2888. <https://doi.org/10.1038/s41467-018-05126-3>.
- McCain, C.M., King, S.R.B., Szcwzyk, T., Beck, J., 2018. Small mammal species richness is directly linked to regional productivity, but decoupled from food resources, abundance, or habitat complexity. *J. Biogeogr.* 45, 2533–2545. <https://doi.org/10.1111/jbi.13432>.
- Mittermeier, R.A., Wilson, D.E., Rylands, A.B., 2013. *Handbook of the Mammals of the World*. Retrieved from, Lynx Edicions, Barcelona, Spain.
- Montaño-Centellas, F., Baiser, B., McGrew, A., Trotta, L., Li, D., 2022. Global patterns and drivers of raptor phylogenetic and functional diversity. *Glob. Ecol. Biogeogr.* 32, 281–294. <https://doi.org/10.1111/geb.13619>.
- Montaño-Centellas, F.A., McCain, C., Loiselle, B.A., Grytnes, J.-A., 2020. Using functional and phylogenetic diversity to infer avian community assembly along elevational gradients. *Glob. Ecol. Biogeogr.* 29 (2), 232–245. <https://doi.org/10.1111/geb.13021>.
- Mukeka, J.M., Ogutu, J.O., Kanga, E., Roskaf, E., 2019. Human-wildlife conflicts and their correlates in Narok County, Kenya. *Glob. Ecol. Conserv.* 18, e00620 <https://doi.org/10.1016/j.gecco.2019.e00620>.
- Murali, G., Gumbs, R., Meiri, S., Roll, U., 2021. Global determinants and conservation of evolutionary and geographic rarity in land vertebrates. *Sci. Adv.* 7, eabe5582 <https://doi.org/10.1126/sciadv.abe5582>.
- Musila, S., Monadjem, A., Webala, P.W., Patterson, B.D., Hutterer, R., De Jong, Y.A., Butynski, T.M., Mwambi, G., Chen, Z.Z., Jiang, X.L., 2019. An annotated checklist of mammals of Kenya. *Zool. Res.* 40, 3–52. <https://doi.org/10.24272/j.issn.2095-8137.2018.059>.
- NASA JPL, 2020. NASADEM Merged DEM Global 1 arc second V001, in: NASA EOSDIS Land Processes DAAC (Ed.). Retrieved from https://doi.org/10.5067/MEASURES/NASADEM/NASADEM_HGT.001.
- Neugarten, R.A., Moull, K., Martinez, N.A., Andriamaro, L., Bernard, C., Bonham, C., Cano, C.A., Ceotto, P., Cutter, P., Farrell, T.A., Gibb, M., Goedschalk, J., Hole, D., Honzák, M., Kasecker, T., Koenig, K., Larsen, T.H., Ledezma, J.C., McKinnon, M., Mulligan, M., Nijbroek, R., Olsson, A., Rakotobe, Z.L., Rasolohery, A., Saenz, L., Steininger, M., Wright, T.M., Turner, W., 2020. Trends in protected area representation of biodiversity and ecosystem services in five tropical countries. *Ecosyst. Serv.* 42, 101078 <https://doi.org/10.1016/j.ecoser.2020.101078>.
- Obersoler, V., Tenan, S., Zipkin, E.F., Rovero, F., 2020. Poor management in protected areas is associated with lowered tropical mammal diversity. *Anim. Conserv.* 23, 171–181. <https://doi.org/10.1111/acv.12525>.
- Ogutu, J.O., Piepho, H.-P., Said, M.Y., Ojwang, G.O., Njino, L.W., Kifugo, S.C., Wargute, P.W., Paiva, S.R., 2016. Extreme wildlife declines and concurrent increase in livestock numbers in Kenya: What are the causes? *PLoS One* 11 (9), e0163249. <https://doi.org/10.1371/journal.pone.0163249>.
- Oliveira, B.F., Machac, A., Costa, G.C., Brooks, T.M., Davidson, A.D., Rondinini, C., Graham, C.H., 2016. Species and functional diversity accumulate differently in mammals. *Glob. Ecol. Biogeogr.* 25, 1119–1130. <https://doi.org/10.1111/geb.12471>.

- Onditi, K.O., Li, X.-Y., Song, W.-Y., Li, Q., Musila, S., Mathenge, J., Kioko, E., Jiang, X.-L., 2021. The management effectiveness of protected areas in Kenya. *Biodivers. Conserv.* 30, 3813–3836. <https://doi.org/10.1007/s10531-021-02276-7>.
- Onditi, K.O., Song, W.-Y., Li, X.-Y., Chen, Z.-Z., Li, Q., He, S.-W., Musila, S., Kioko, E., Jiang, X.-L., 2022. Patterns and predictors of small mammal phylogenetic and functional diversity in contrasting elevational gradients in Kenya. *Front. Ecol. Evol.* 9, 742524. <https://doi.org/10.3389/fevo.2021.742524>.
- Pavoine, S., Bonsall, M.B., 2011. Measuring biodiversity to explain community assembly: a unified approach. *Biol. Rev. Camb. Philos. Soc.* 86, 792–812. <https://doi.org/10.1111/j.1469-185X.2010.00171.x>.
- Philip, P., Peter, O., Rob, M., 2015. AFRICLIM 3.0: high-resolution ensemble climate projections for Africa, Figshare. Retrieved from <https://webfiles.york.ac.uk/KITE/AfricClim/>.
- Qian, H., 2009. Global tests of regional effect on species richness of vascular plants and terrestrial vertebrates. *Ecography* 32, 553–560. <https://doi.org/10.1111/j.1600-0587.2008.05755.x>.
- Rada, S., Schweiger, O., Harpke, A., Kühn, E., Kuras, T., Settele, J., Musche, M., Andersen, A., 2019. Protected areas do not mitigate biodiversity declines: A case study on butterflies. *Divers. Distrib.* 25, 217–224. <https://doi.org/10.1111/ddi.12854>.
- Rahbek, C., 1995. The elevational gradient of species richness: a uniform pattern? *Ecography* 18, 200–205. <https://doi.org/10.1111/j.1600-0587.1995.tb00341.x>.
- Ramesh, T., Kalle, R., Downs, C.T., 2016. Predictors of mammal species richness in KwaZulu-Natal, South Africa. *Ecol. Ind.* 60, 385–393. <https://doi.org/10.1016/j.ecolind.2015.07.011>.
- Ricklefs, R.E., 2004. A comprehensive framework for global patterns in biodiversity. *Ecol. Lett.* 7, 1–15. <https://doi.org/10.1046/j.1461-0248.2003.00554.x>.
- Rosenzweig, M.L., 1992. Species diversity gradients: We know more and less than we thought. *J. Mammal.* 73, 715–730. <https://doi.org/10.2307/1382191>.
- Rosenzweig, M.L., 1995. *Species Diversity in Space and Time*. Cambridge University Press, Cambridge, United Kingdom.
- Rowan, J., Beaudrot, L., Franklin, J., Reed, K.E., Smail, I.E., Zamora, A., Kamilar, J.M., 2020. Geographically divergent evolutionary and ecological legacies shape mammal biodiversity in the global tropics and subtropics. *Proc. Natl. Acad. Sci. U.S.A.* 117 (3), 1559–1565. <https://doi.org/10.1073/pnas.1910489116>.
- Rowe, R.J., 2009. Environmental and geometric drivers of small mammal diversity along elevational gradients in Utah. *Ecography* 32, 411–422. <https://doi.org/10.1111/j.1600-0587.2008.05538.x>.
- Running, S., Mu, Q., Zhao, M., Moreno, A., 2019. MOD16A3GF MODIS/Terra Net Evapotranspiration Gap-Filled Yearly L4 Global 500 m SIN Grid V006, NASA EOSDIS Land Processes DAAC. Retrieved from <https://doi.org/10.5067/MODIS/MOD16A3GF.006>.
- Safi, K., Cianciaruso, M.V., Loyola, R.D., Brito, D., Armour-Marshall, K., Diniz-Filho, J.A., 2011. Understanding global patterns of mammalian functional and phylogenetic diversity. *Philos. Trans. R. Soc.* B 366, 2536–2544. <https://doi.org/10.1098/rstb.2011.0024>.
- Shi, H., Singh, A., Kant, S., Zhu, Z., Waller, E., 2005. Integrating habitat status, human population pressure, and protection status into biodiversity conservation priority setting. *Conserv. Biol.* 19, 1273–1285. <https://doi.org/10.1111/j.1523-1739.2005.00225.x>.
- Song, W.-Y., Li, X.-Y., Chen, Z.-Z., Li, Q., Onditi, K.O., He, S.-W., Jiang, X.-L., 2020. Isolated alpine habitats reveal disparate ecological drivers of taxonomic and functional beta-diversity of small mammal assemblages. *Zool. Res.* 41, 670–683. <https://doi.org/10.24272/j.issn.2095-8137.2020.085>.
- Stein, A., Gerstner, K., Kreft, H., Arita, H., 2014. Environmental heterogeneity as a universal driver of species richness across taxa, biomes and spatial scales. *Ecol. Lett.* 17 (7), 866–880. <https://doi.org/10.1111/ele.12277>.
- Stein, A., Kreft, H., 2015. Terminology and quantification of environmental heterogeneity in species-richness research. *Biol. Rev. Camb. Philos. Soc.* 90, 815–836. <https://doi.org/10.1111/brv.12135>.
- Theodoridis, S., Fordham, D.A., Brown, S.C., Li, S., Rahbek, C., Nogues-Bravo, D., 2020. Evolutionary history and past climate change shape the distribution of genetic diversity in terrestrial mammals. *Nat. Commun.* 11, 2557. <https://doi.org/10.1038/s41467-020-16449-5>.
- Thomsen, M.S., Altieri, A.H., Angelini, C., Bishop, M.J., Bulleri, F., Farhan, R., Frühling, V.M.M., Gribben, P.E., Harrison, S.B., He, Q., Klinghardt, M., Langeneck, J., Lanham, B.S., Mondardini, L., Mulders, Y., Oleksyn, S., Ramus, A.P., Schiel, D.R., Schneider, T., Siciliano, A., Silliman, B.R., Smale, D.A., South, P.M., Wernberg, T., Zhang, S., Zotz, G., 2022. Heterogeneity within and among co-occurring foundation species increases biodiversity. *Nat. Commun.* 13, 581. <https://doi.org/10.1038/s41467-022-28194-y>.
- Thuiller, W., Broennimann, O., Hughes, G., Alkemade, J.R.M., Midgley, G.F., Corsi, F., 2006. Vulnerability of African mammals to anthropogenic climate change under conservative land transformation assumptions. *Glob. Chang. Biol.* 12, 424–440. <https://doi.org/10.1111/j.1365-2486.2006.01115.x>.
- Tucker, C.M., Cadotte, M.W., Carvalho, S.B., Davies, T.J., Ferrier, S., Fritz, S.A., Grenyer, R., Helmus, M.R., Jin, L.S., Mooers, A.O., Pavoine, S., Purschke, O., Redding, D.W., Rosauer, D.F., Winter, M., Mazel, F., 2017. A guide to phylogenetic metrics for conservation, community ecology and macroecology. *Biol. Rev. Camb. Philos. Soc.* 92, 698–715. <https://doi.org/10.1111/brv.12252>.
- Tyrrell, P., du Toit, J., Macdonald, D., 2020. Conservation beyond protected areas: Using vertebrate species ranges and biodiversity importance scores to inform policy for an east African country in transition. *Conserv. Sci. Pract.* 2, e136. <https://doi.org/10.1111/csp.2.136>.
- UNEP-WCMC, IUCN, 2021. Protected Planet: The World Database on Protected Areas (WDPA) and World Database on Other Effective Area-based Conservation Measures (WD-OECM) [Online]. UNEP-WCMC and IUCN, Cambridge, UK. Retrieved from www.protectedplanet.net.
- Veldhuis, M.P., Ritchie, M.E., Ogutu, J.O., Morrison, T.A., Beale, C.M., Estes, A.B., Mwakilema, W., Ojwang, G.O., Parr, C.L., Probert, J., Wargute, P.W., Hopcraft, J.G. C., Olff, H., 2019. Cross-boundary human impacts compromise the Serengeti-Mara ecosystem. *Science* 363, 1424–1428. <https://doi.org/10.1126/science.aav0564>.
- Venter, O., Sanderson, E.W., Magrath, A., Allan, J.R., Beher, J., Jones, K.R., Possingham, H.P., Laurance, W.F., Wood, P., Fekete, B.M., Levy, M.A., Watson, J.E., 2018. Last of the Wild Project, Version 3 (LWP-3): 2009 Human Footprint, 2018 Release. NASA Socioeconomic Data and Applications Center (SEDAC), Palisades, NY. Retrieved from <https://doi.org/10.7927/H46T0JQ4>.
- Venter, O., Possingham, H.P., Watson, J.E.M., 2020. The human footprint represents observable human pressures: Reply to Kennedy et al. *Glob. Change Biol.* 26, 330–332. doi: 10.1111/gcb.14849.
- Venter, O., Sanderson, E.W., Magrath, A., Allan, J.R., Beher, J., Jones, K.R., Possingham, H.P., Laurance, W.F., Wood, P., Fekete, B.M., Levy, M.A., Watson, J.E., 2016. Sixteen years of change in the global terrestrial human footprint and implications for biodiversity conservation. *Nat. Commun.* 7, 12558. <https://doi.org/10.1038/ncomms12558>.
- Vetaas, O.R., 2006. Biological relativity to water–energy dynamics: a potential unifying theory? *J. Biogeogr.* 33, 1866–1867. <https://doi.org/10.1111/j.1365-2699.2006.01618.x>.
- Webb, C.O., Ackerly, D.D., McPeck, M.A., Donoghue, M.J., 2002. Phylogenies and community ecology. *Annu. Rev. Ecol. Syst.* 33, 475–505. <https://doi.org/10.1146/annurev.ecolsys.33.010802.150448>.
- Webb, C.O., Donoghue, M.J., 2005. Phylomatic: tree assembly for applied phylogenetics. *Mol. Ecol. Notes* 5, 181–183. <https://doi.org/10.1111/j.1471-8286.2004.00829.x>.
- Weiher, E., Clarke, G.D.P., Keddy, P.A., 1998. Community assembly rules, morphological dispersion, and the coexistence of plant species. *Oikos* 81, 309–322. <https://doi.org/10.2307/3547051>.
- Wen, Z., Cai, T., Feijó, A., Xia, L., Cheng, J., Ge, D., Yang, Q., 2020. Using completeness and defaunation indices to understand nature reserve's key attributes in preserving medium- and large-bodied mammals. *Biol. Conserv.* 241, 108273. <https://doi.org/10.1016/j.biocon.2019.108273>.
- Western, D., Musyoki, C., Mwangi, E., Mwachala, G., Said, M., Wargute, P., Matiku, P., Landsberg, F., Kamala, E., Waruingi, L., Kariuki, P., Situma, C., Ojwang, G., Njino, L., Mullenke, L., Muli, D., Malombe, I., Marchant, R., Platts, P., Stickler, M., 2015. Kenya's Natural Capital: A Biodiversity Atlas. Ministry of Environment Natural Resources and Regional Development Authorities, Kenya, Nairobi, Kenya.
- Wilman, H., Belmaker, J., Simpson, J., de la Rosa, C., Rivadeneira, M.M., Jetz, W., 2014. EltonTraits 1.0: Species-level foraging attributes of the world's birds and mammals. *Ecology* 95, 2027. <https://doi.org/10.1890/13-1917.1>.
- Woodbridge, J., Fyfe, R., Smith, D., Pelling, R., Vareilles, A., Batchelor, R., Bevan, A., Davies, A.L., Lafontaine, G., 2021. What drives biodiversity patterns? Using long-term multidisciplinary data to discern centennial-scale change. *J. Ecol.* 109 (3), 1396–1410. <https://doi.org/10.1111/1365-2745.13565>.
- Zhang, Q., Ye, J.F., Le, C.T., Njenga, D.M., Rabarijaona, N.R., Omollo, W.O., Lu, L.M., Liu, B., Chen, Z.D., 2022. New insights into the formation of biodiversity hotspots of the Kenyan flora. *Divers. Distrib.* 28, 2696–2711. <https://doi.org/10.1111/ddi.13624>.
- Zomer, R., Trabucco, A., 2022. Global Aridity Index and Potential Evapotranspiration (ETO) Database: Version 3. Retrieved from <https://doi.org/10.6084/m9.figshare.7504448.v5>.
- Zupan, L., Cabeza, M., Maiorano, L., Roquet, C., Devictor, V., Lavergne, S., Mouillot, D., Mouquet, N., Renaud, J., Thuiller, W., Loyola, R., 2014. Spatial mismatch of phylogenetic diversity across three vertebrate groups and protected areas in Europe. *Divers. Distrib.* 20 (6), 674–685. <https://doi.org/10.1111/ddi.12186>.
- Zurell, D., 2020. mecofun: useful functions for macroecology and species distribution modelling. R Package version 0.5.1. Retrieved from <https://gitup.uni-potsdam.de/macroecology/mecofun>.